

Discussion Paper

Migration and the New Zealand economy

Evidence from sign-restricted SVARs

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Abstract

We investigate the impact of expansionary immigration shocks on the New Zealand economy using sign-restricted structural vector autoregression (SVAR) models. Our key findings indicate that an expansionary immigration shock results in a statistically significant increase (at the 68% credible interval) in the level of house prices and household credit relative to the baseline, while the unemployment rate shows a significant decrease. Moreover, labour productivity and headline consumer price levels show no significant deviation from baseline. However, non-tradable price levels (a sub-component of headline) show a statistically significant increase. Our variance decomposition analysis shows that an immigration shock is a major contributor towards the variation in real GDP levels (30%) and the unemployment rate (30%), at the three year horizon. Further, the contribution of the participation rate (25%) and house price levels (20%) is non-negligible after three years. Our robustness analysis based on immigration shocks for “younger” and “older” age groups, indicates that the increase in “younger” immigrants is associated with a statistically significant decrease in the aggregate unemployment rate, while “older” immigrants are associated with increased labour productivity.

JEL Classifications: C11, C32, F22, F41, F66, J21

Keywords: Immigration shocks, Labour, Housing, Small open economy, Structural Vector Autoregression (SVAR).

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1 Introduction

What are the effects of migration on New Zealand's macroeconomy? This question is an important consideration for policymakers in New Zealand given the increasing share of immigrants as a percentage of the total population, rising from 15% in 1990 to 27% in 2019. This proportion exceeds that of most OECD countries, including Canada (21%), the United States (14%), Sweden (16%), Germany (14%), Norway (14%), the United Kingdom (13%), and France (12%). Among comparable economies, only Australia (29%) has a higher immigrant share.¹ Theoretically, migration can have positive long-term impacts on the New Zealand macroeconomy, increasing GDP, labour supply and house prices. However, realising sustained benefits from an expansionary immigration shock may require managing several short-term adjustments. On the supply side, rapid population growth may strain housing supply, placing upward pressure on house prices and rents. On the demand side, increased consumption may generate inflationary pressure (which may be partially offset by an exchange rate appreciation that reduces import prices and improves terms of trade). In this regard, New Zealand is an interesting case study as it has seen a stronger net migration growth compared to the growth of the New Zealand-born population from 2013 to 2019 (Figure 1), where increases from net migration are more than double the size of natural increase in some years.² Therefore, understanding the impacts of migration is important for policymakers and researchers.

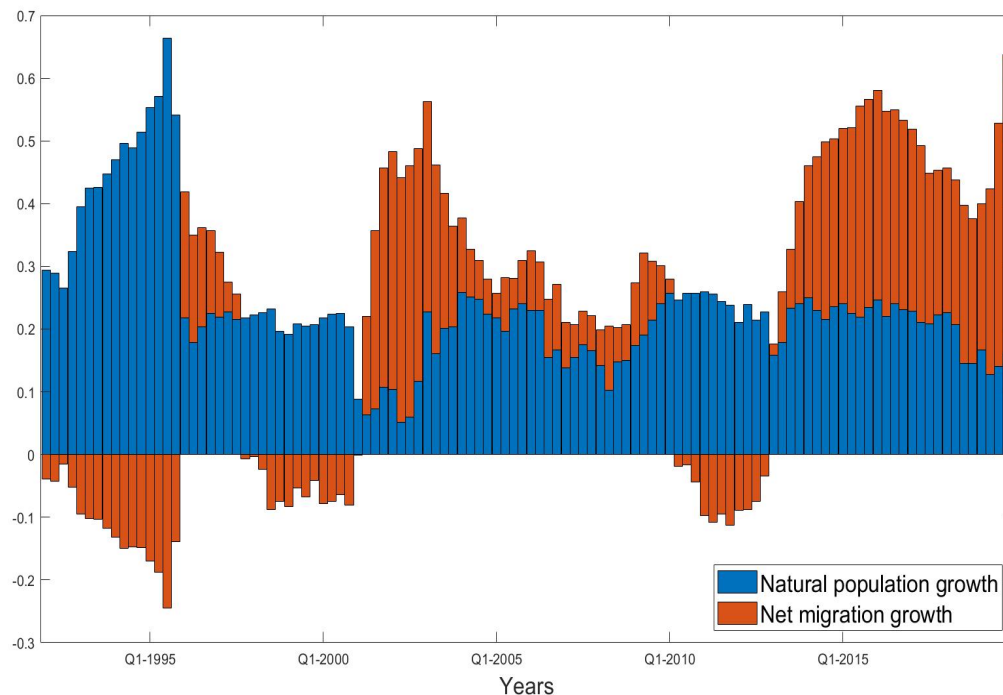
Although assessing the impact of migration is important, research on this topic is scarce for New Zealand. Our paper addresses this gap by examining the effects of an expansionary immigration shock on the New Zealand macroeconomy using sign-restricted structural vector autoregression (SVAR) models. We follow the identification approach of Furlanetto and Robstad (2019), imposing sign restrictions on the impact responses of impulse response functions (IRFs). Our identification strategy achieves full identification of all five structural shocks, ensuring that the estimated immigration shock is separated from other labour market or business cycle disturbances.³ Specifically, we identify five shocks: immigration (proxy for foreign labour supply), domestic labour supply, business cycle, technology, and residual shocks. Our estimation data includes real GDP, real wages, the labour market participation rate, and the unemployment rate. Furthermore, we measure the impact of immigration shocks on consumer and house price levels, household credit levels, labour productivity, the Official Cash Rate (OCR), and the real exchange rate.

¹Source: World Bank (n.d.).

²Stillman and Maré (2008) note for New Zealand: "international migration flows are an important contributor to population fluctuations".

³Furlanetto and Robstad (2019) apply this identification scheme to quantify the effects of immigration shocks on the Norwegian economy.

Figure 1: Quarterly population change (%) for New Zealand, 1992q1-2019q4.



Source: Stats NZ and authors' calculations.

Our paper contributes to the New Zealand literature in two distinct ways. First, we examine the impact of an expansionary immigration shock on the New Zealand economy using the sign-restricted SVAR model of Furlanetto and Robstad (2019) estimated with the Bayesian techniques of Arias et al. (2018). Our baseline findings indicate that the unemployment rate falls in response to an expansionary immigration shock, while the level of house prices rise. Headline consumer price levels respond insignificantly except for a statistically significant increase (at the 68% credible interval) in non-tradable price levels. Our forecast error variance decomposition (FEVD) analysis shows that an immigration shock is a major contributor at the three year horizon towards real GDP levels (30%) and the unemployment rate (30%), and a non-negligible driver of the participation rate (25%) and house price levels (20%).⁴

Second, we compare the impact of a “younger” immigration shock to an “older” immigration shock. Immigrants are split into two age groups: “younger” immigrants aged 15-29 years and

⁴We specify the contribution of above 25% as ‘major’, 20% to 25% as ‘non-negligible’ and below 20% as ‘negligible’.

“older” immigrants aged 30-44 years.⁵ Our results show a significant and pronounced fall in the unemployment rate associated with the “younger” immigration shock, while the “older” immigration shock is associated with an increase in labour productivity.

The rest of the paper is structured as follows. Section 2 presents the related literature. Section 3 describes the data, model and the identification scheme. Section 4 presents our main results. Section 5 contains robustness checks, and Section 6 concludes.

2 Related Literature

The migration literature has few applications of SVAR modelling. Recent and related contributions are: Schiman (2021) for Austria, Kiguchi and Mountford (2019), Weiske (2019) for the US, Furlanetto and Robstad (2019) for Norway and Smith and Thoenissen (2019), Armstrong and McDonald (2016), Vehbi (2016), McDonald (2013) for New Zealand.

Our work is closely related to Furlanetto and Robstad (2019), who use a sign-restricted SVAR model for Norway estimated from 1990 $q1$ to 2014 $q2$. Their sign restrictions follow from a DSGE model of the labour market with search and matching frictions (Furlanetto & Robstad, 2018), which assume that immigrants rapidly enter the Norwegian labour market.⁶ They include only the net flow of workers migrating from EU/EFTA countries, North America, Australia, New Zealand, and eastern Europe derived from the net immigration series provided by Statistics Norway and the unemployment rate from the Norwegian Labor and Welfare Administration. Additional annual data from Statistics Norway confirm that work is the major motivation for immigration from the countries included in their analysis. Our assumption that immigrants enter employment rapidly is supported by New Zealand’s points-based migration system, which selects for skilled workers with high employability.⁷ Furlanetto and Robstad (2019) find that in response to an expansionary immigration shock the unemployment rate decreases significantly, more for natives than for foreign workers. Interestingly, they find a negative impact on productivity and no significant impact on house price levels. Along the same lines, Schiman (2021) applies the sign-restricted identification scheme of Foroni et al. (2018) in an SVAR model to study the impact of positive foreign labour supply shocks associated with the free movement of workers from eastern Europe to Austria. He finds a significant increase

⁵We study the impact of “younger” and “older” immigration shocks on the same aggregate macroeconomic variables as in our baseline model.

⁶Furlanetto and Robstad (2018) extends Foroni et al. (2018) by incorporating migration into the model.

⁷Notable exceptions to the point-based system includes immigrants from Australia, UK working holiday visa and Pacific Islands, among others.

in the log-levels of real GDP, a temporary increase in the inflation rate and the unemployment rate, and a fall in the log-levels of real wages in response to the foreign labour supply shocks for Austria.

New Zealand's literature on migration focuses primarily on the housing market. In one of the first applications for New Zealand, Coleman and Landon-Lane (2007) use an SVAR model to show that a net migration increase equivalent to 1 percentage points rise in population is associated with a 10 percentage points increase in house prices. Stillman and Maré (2008) apply microeconomic techniques to examine the impact of population changes on house prices, via international and internal migration. They find that a 1% increase in inflows of returning New Zealanders leads to a 6 to 9% increase in house prices, while they find no evidence of a positive impact on house prices in response to foreign-born migrant inflows. McDonald (2013) investigates the asymmetric impact of migrant arrivals and departures using Cholesky identified SVARs. He finds that migrant arrivals have a more substantial impact on house prices than migrant departures. A permanent 1000-person increase in monthly arrivals raises real house prices by 4%, whereas a 1000-person fall in monthly departures raises real house prices by only 2%. Armstrong and McDonald (2016) investigate the labour market response to an increase in net migration using a Cholesky identified SVAR. Their findings indicate that the increase in New Zealand's net migration from 2013-2015 is largely driven by a higher Australian unemployment rate. Vehbi (2016) extends the work of Armstrong and McDonald (2016) and finds that the inflow of migrants aged 30-49 years has a relatively larger impact on the annual percentage changes of consumption, real house prices, real rents, and residential investment compared to the inflow of 17-29 year old migrants. Another study by Smith and Thoenissen (2019) employ a Cholesky identified SVAR model and find that a migration shock contributes more towards volatility in per-capita GDP (19%), investment (17%) and consumption (12%) compared to real house prices (4%) and residential investment (3%).⁸

We contribute to the New Zealand literature by employing sign-restricted SVAR models to examine how the New Zealand macroeconomy responds to an expansionary immigration shock. Our main findings display a persistent fall in the unemployment rate, with a corresponding hump-shaped response in the immigration rate, real GDP level, and the participation rate. We find no evidence of significant movements in price levels, except for a statistically significant increase (at the 68% credible interval) in non-tradable prices. Moreover, we find that the fall in the aggregate unemployment rate is more pronounced in response to a "younger" immigration shock compared to that of an "older" immigration shock.

In regard to international literature, Kiguchi and Mountford (2019) evaluate the impact of migration on the US economy from 1950 to 2005 using annual data. Their sign restrictions for an

⁸Smith and Thoenissen (2019) find similar results through an estimated DSGE model. They report a positive effect on per-capita GDP and its components when the average immigrant is more skilled than locals.

eight-variable SVAR are based on a DSGE model that incorporates the labour market with search and matching frictions similar to that of Furlanetto and Robstad (2019).⁹ Kiguchi and Mountford (2019) find a temporary increase in the log-levels of unemployment per capita following an increase in net migration, while the log-levels of real GDP per capita and real consumption per capita fall on impact before a subsequent recovery.

Weiske (2019) also examines the impact of an increase in net migration on the US economy using quarterly data from 1959 to 2016 in an SVAR model identified through long-run restrictions. The three shocks identified in his model are: investment technology, neutral technology and immigration shocks. In Weiske's model, the response of immigration to technology shocks implies that long-run economic conditions play a role in the decision of whether immigrants choose to settle in US. He further finds that the log-levels of GDP per capita, consumption per capita, and hours worked per capita barely respond to an immigration shock. After 10 quarters, however, the response of GDP per capita becomes significantly positive. Real wages decrease significantly for a period of two years, whereas investment per capita increases significantly for five years. Overall, Weiske finds immigration shocks to be a negligible contributor to the US economy, accounting for only 10% of fluctuations in GDP, consumption, investment and wages.¹⁰

Ottaviano and Peri (2008) use the education-experience group approach of Borjas (2003) and find a significant degree of imperfect substitution between immigrants and natives. Peri and Sparber (2009) report similar findings. In their setting, immigrants specialise in manufacturing tasks, while natives fill in communication intensive jobs. As a result, migration-induced wage reductions are significantly mitigated. Peri and Sparber (2009) further show that the higher supply of manufacturing work in American states with large migration inflows "more than compensated for the change in skill supply among natives", which is associated with higher wages in communication intensive occupations. In contrast, Dustmann et al. (2005) find no evidence of adverse labour market effects between 1983 and 2000 for the UK, even though the skills and education levels of immigrants and natives are similar.

⁹Kiguchi and Mountford (2019) identify only a business cycle shock and a migration shock out of the eight structural shocks. There is a shock masquerading problem if the number of identified shocks is less than the number of variables (Fry & Pagan, 2011; Wolf, 2020).

¹⁰See d'Albis et al. (2016) for a study of migration in France using a Cholesky identified SVAR model.

3 Methodology

3.1 Data

We use quarterly data from 1992 $q1$ to 2019 $q4$.¹¹ The starting date corresponds broadly to the adoption of inflation targeting in New Zealand. The end date marks the onset of the COVID-19 pandemic as including the sequence of extreme observations during the COVID-19 period may significantly affect parameter estimates (Lenza & Primiceri, 2022). Moreover, migration trends changed considerably in response to the COVID-19 public health measures, which is another reason for limiting our sample to 2019.

The variables in our baseline model include the immigration rate (ir), real GDP levels (y), real wage levels (w), the participation rate (pr), and the unemployment rate ($unemp$). Following Furlanetto and Robstad (2019), we construct the immigration rate series as a ratio of net migrant stock to working-age population (WAP). The participation rate is defined as the proportion of WAP who are in the labour force. Further, we replace the unemployment rate in our baseline model with the following variables, one at a time: headline consumer price index in levels (CPI), two core consumer price measures in levels, one excludes food and energy while the other excludes government charges as well, consumer price levels for tradeables, non-tradeables and services, levels of house prices and household credit, labour productivity, real exchange rate-trade weighted index (RER-TWI) the policy rate proxied by the official cash rate (OCR) and 3-month interbank rate. We leave the responses of our variables of interest such as the unemployment rate and various measures of consumer price levels unrestricted to let the data speak, whereas variables such as real GDP, real wages and, the participation rate are used to identify the shocks in the model. All variables are expressed in log-levels except the unemployment rate and the policy rate, which are expressed in percentage points. Consequently, the IRFs show the percentage deviations from baseline for the variables in log-levels, while the unemployment rate and policy rate responses show the absolute deviations. This specification means, for example, that a positive response of consumer prices indicates a higher price level relative to baseline, not necessarily higher inflation.

3.2 Model

Our structural model is given by:

$$\mathbf{y}'_t \mathbf{A}_0 = \sum_{l=1}^p \mathbf{y}'_{t-l} \mathbf{A}_l + \mathbf{c}' + \boldsymbol{\epsilon}'_t, \quad (1)$$

¹¹ Data description and sources are in Appendix A.

where $\mathbf{y}'_t = [ir_t, y_t, w_t, pr_t, unemp_t]$. $\mathbf{y}$ is a $(n \times 1)$ vector of variables, with n denoting the total number of variables. Similarly, ϵ_t is the vector of structural shocks. $\mathbf{A}_i$, for $0 \leq i \leq p$, are $(n \times n)$ matrices of structural parameters, with $\mathbf{A}_0$ invertible. $\mathbf{c}$ is a $(n \times 1)$ vector of constants, $p=2$ is the lag length, and T is the sample size. Conditional on past information and initial conditions $\mathbf{y}_0, \dots, \mathbf{y}_{1-p}$, the vector ϵ_t is Gaussian with mean zero and covariance matrix $\mathbb{I}_n$. Following Rubio-Ramirez et al. (2010), we can write the SVAR in compact form:

$$\mathbf{y}'_t \mathbf{A}_0 = \mathbf{x}'_t \mathbf{A}_+ + \epsilon'_t, \quad (2)$$

where $\mathbf{x}'_t = [\mathbf{y}'_{t-1} \dots \mathbf{y}'_{t-p} \ 1]$ for $1 \leq t \leq T$. $\mathbf{A}_0$ and $\mathbf{A}_+ = [\mathbf{A}'_1 \dots \mathbf{A}'_p \ \mathbf{c}']$ are matrices of structural parameters.

Post-multiplying Equation (2) by $\mathbf{A}_0^{-1}$, we obtain the reduced-form Vector Autoregression (VAR) model:

$$\mathbf{y}'_t = \mathbf{x}'_t \mathbf{B} + u'_t, \quad (3)$$

where $\mathbf{B} = \mathbf{A}_+ \mathbf{A}_0^{-1}$, $u'_t = \epsilon'_t \mathbf{A}_0^{-1}$ and $\mathbf{E}[u_t u'_t] = \Sigma = (\mathbf{A}_0 \mathbf{A}'_0)^{-1}$. $\mathbf{B}$ is the matrix of reduced-form coefficients and Σ is the residual variance-covariance matrix.

3.3 Identification

We adapt the identification scheme of Furlanetto and Robstad (2019) to the New Zealand context which allows us to fully identify our five-variable SVAR model with five shocks. These shocks are: a business cycle shock, two labour market shocks: immigration and domestic labour supply, a technology shock, and a residual shock. Our restrictions are heavily grounded in the New Keynesian model of Foroni et al. (2018), which is extended to include immigration shocks as modeled in Kiguchi and Mountford (2019), Lozej (2019) and Weiske (2019). The major advantage of our fully identified model is that it successfully disentangles the immigration shock, which is the main focus of this study, from the rest of the shocks in the system. Our identification restrictions, imposed only on impact, are summarised in Table 1. We implement the sign restrictions using the algorithm of Arias et al. (2018), which draws from the posterior distribution of structural parameters and retains only those draws that satisfy our sign restrictions.

We define an expansionary immigration shock as an exogenous increase in net migration that raises the immigration rate (the ratio of migrant stock to working-age population). By construction, this shock causes a positive co-movement between real GDP and the participation rate, as well as a negative co-movement between real GDP and real wages. Based on the points-based migration system of New Zealand, most immigrants are assumed to be highly skilled and are likely to enter

Table 1: Impact restrictions in the baseline model

<i>Variables</i>	<i>Shocks</i>				
	Immigration	Business cycle	Technology	Domestic labour supply	Residual
Immigrants/participants	+	NA	NA	-	NA
Real GDP	+	+	+	+	+
Real wages	-	+	+	-	-
Participation rate	+	+	-	+	-
Unemployment rate	NA	NA	NA	NA	NA

Note: +(-) means a positive(negative) sign-restriction imposed on impact on IRFs. NA means no restrictions. We impose two additional magnitude restrictions along with the sign restrictions that allow us to disentangle immigration shocks from domestic labour supply shocks: (i) In response to an expansionary immigration shock, the impact response of the immigration rate is positive and the impact response of the participation rate is positive and less than the impact response of the immigration rate, and (ii) In response to an expansionary domestic labour supply shock, the impact response of the participation rate is positive and the impact response of the immigration rate is either positive and less than the impact response of the participation rate or it is negative. This combination of sign and magnitude restrictions allow us to impose restrictions on the ratio of immigrants to participants, which otherwise is unobserved in our model.

the labour market relatively quickly, which generates economic activity. Moreover, rapid inclusion of immigrants into the labour force is associated with an increase in the participation rate. Thus, the response of the participation rate is restricted to be positive on impact to an expansionary immigration shock. The response of real wages are restricted to be negative on impact to an expansionary immigration shock following Furlanetto and Robstad (2018), Galí et al. (2011), and Ottaviano and Peri (2008, 2012) and Borjas (2003), who show that a foreign labour supply shock yields a negative short-term response of real wages. Notably, both the SVAR and DSGE models of Smith and Thoenissen (2019) fitted to New Zealand data show a short-term decrease in log-levels of detrended real wages in response to a positive net migration shock.

Since the domestic labour force consists of both native and foreign-born workers, the IRFs of real GDP, real wages, and the participation rate should exhibit a similar response to either an immigration or domestic labour supply shock.¹² Therefore, we differentiate the domestic labour supply shock from an immigration shock by imposing sign restrictions on the ratio of immigrants to partici-

¹²IRFs of domestic labour supply shocks are in Appendix B.

pants, following Furlanetto and Robstad (2019). By construction, an expansionary domestic labour supply shock is assumed to decrease the ratio of immigrants to participants, which implies that the inflow of native workers into the labour market is larger than any subsequent increase in net migration. Similarly, an expansionary immigration shock will, by construction, increase the ratio of immigrants to participants. This assumes that immigrants are quickly absorbed into the labour market and that the increase in the ratio of immigrants to participants is more than the increase in the participation rate of natives. However, note that we do not directly observe the ratio of immigrants to participants in our model.

Both the participation rate and the immigration rate are specified as a proportion of the working-age population (WAP), and hence relative changes in the participation rate and the immigration rate imply sign restrictions on the ratio of immigrants to participants. This is achieved in our model by imposing a combination of sign and magnitude restrictions on impact. First, in response to an expansionary immigration shock, we impose restrictions that ensure a positive impact on both the immigration rate and the participation rate, with the magnitude of the participation rate less than the immigration rate. Second, in response to an expansionary domestic labour supply shock, we impose restrictions that ensure a positive impact on the participation rate, while the impact on the immigration rate is either less than the increase in the participation rate or it is negative.¹³ Hence, the two magnitude restrictions on the immigration rate and the participation rate are crucial in disentangling the immigration shock from the domestic labour supply shock in our model.

The business cycle shock is characterised by pro-cyclical behaviour of the participation rate and real wages. Hence, we impose the restrictions that real GDP, the participation rate and real wages all increase in response to an expansionary business cycle shock. The technology shock is characterised by counter-cyclical dynamics between output and participation rate but pro-cyclical dynamics between output and wages. Our identification of technology shocks is based on the dynamics generated by technology shocks in the DSGE models by Foroni et al. (2018) and Campolmi and Gnocchi (2016).¹⁴ In our model, the negative impact restriction on the participation rate serves our purpose of differentiating technology shocks from business cycle shocks. The introduction of a technology shock primarily ensures that other labour market shocks are not distorted by other shocks in the system.¹⁵

¹³The impact on the immigration rate may be positive or negative in response to an expansionary domestic labour supply shock.

¹⁴Christiano et al. (2015) argue that the response of the participation rate to technology shocks is model-dependent.

¹⁵Our technology shock corresponds to residual shock of Furlanetto and Robstad (2019).

The fifth and final shock we introduce is an expansionary residual shock, which is identified using a set of sign restrictions different from all the other shocks in the system.¹⁶ Our identification of this fifth shock allows us to achieve full identification of the SVAR system. Thus, the business cycle, technology, domestic labour supply and residual shocks serve the purpose of sharpening our identification of the structural immigration shock, which is the main focus of this paper.

4 Results

4.1 Impulse Response Functions

Figure 2 reports our baseline results for the five variables.¹⁷ The solid lines are the point-wise posterior median responses while the shaded regions are the corresponding 68% equal-tailed posterior probability bands.¹⁸ In line with Furlanetto and Robstad (2019), we find a significant fall in the unemployment rate that lasts around 18 months after the immigration shock. To be more precise, a 0.25% expansionary immigration shock is associated with a significant fall in the unemployment rate of over 0.20 percentage points in the medium-term (i.e., 2-3 years).¹⁹ Furlanetto and Robstad (2019) justify this result by their exclusive focus on job-related migration and because these immigrants are rapidly integrated into the labour market. Our assumption that immigrants are highly skilled and enter employment rapidly is supported by New Zealand's points-based migration system, which selects for skilled workers with high employability. Here, we also emphasise the use of quarterly data that absorb much of the delay for migrant integration into the labour market, which will otherwise be more evident in the high-frequency data. We base this assumption on recent work including: (i) "over 60% of foreign-born workers have tertiary education of some form, highlighting the skilled nature of New Zealand's immigrant workforce" (Alinaghi & Ball, 2025), (ii) "over half of all adults in New Zealand with a tertiary diploma or higher-level qualification were born overseas" (Education Counts, 2024), and (iii) OECD data show that while foreign-born New Zealanders with tertiary qualifications have "the highest employment rate of any OECD country", yet they face a 15% negative earnings gap relative to similarly educated New Zealand-born workers, which is "much

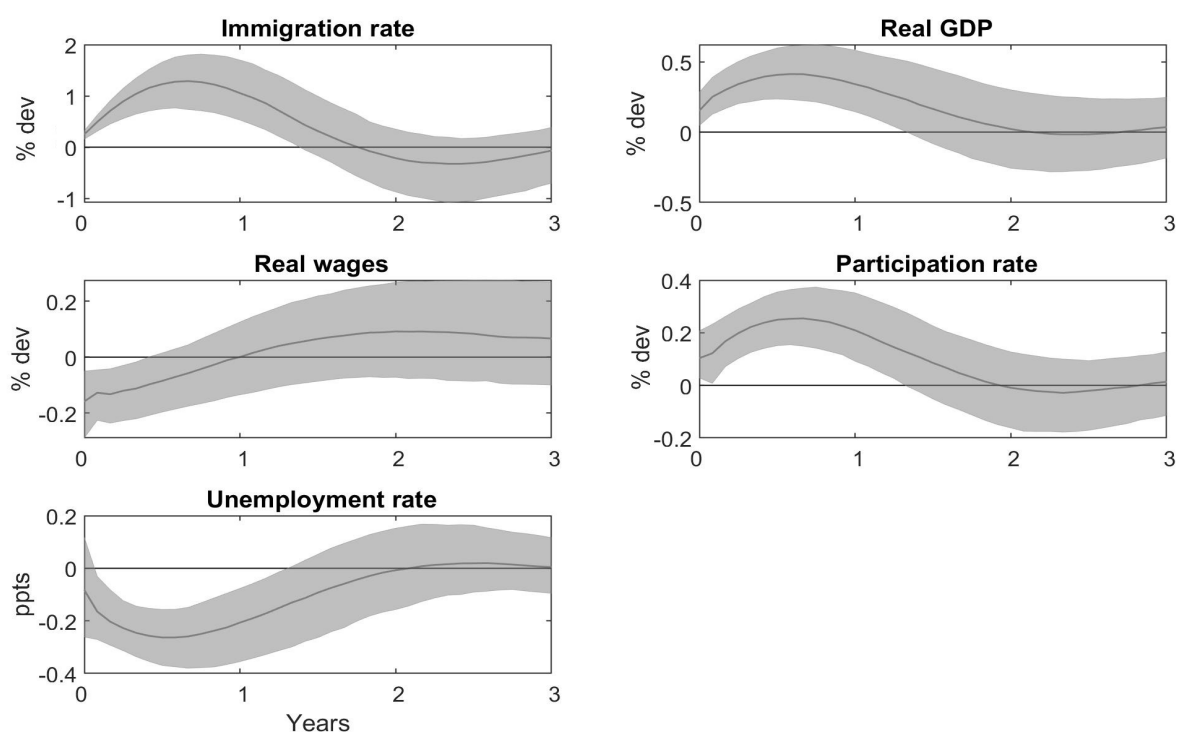
¹⁶Our residual shock corresponds to the wage bargaining shock of Furlanetto and Robstad (2019). Their wage bargaining shock is characterised by counter-cyclical movements of the participation rates and real wages, supported by the DSGE developed in Furlanetto and Robstad (2018). We abstain from identifying a wage bargaining shock, since New Zealand, unlike Norway, does not have a centralized wage negotiation system.

¹⁷IRFs to immigration shocks for all the models are in Appendix B along with some of the IRFs to domestic labour supply shocks.

¹⁸68% probability bands are common in Bayesian SVAR literature (see, for example, Furlanetto and Robstad (2019) and Schiman (2021)).

¹⁹The size of the shock (0.25%) is computed as in R  th and Van der Veken (2023).

Figure 2: Baseline results: IRFs to a 0.25% immigration shocks.



Key insight: The unemployment rate falls significantly in response to an expansionary immigration shock.

larger than that experienced by similarly educated foreign-born workers in other OECD countries” (Education Counts, 2024). Based on these findings, we argue that migrants are integrated into the New Zealand workforce with relative efficiency, although the speed of absorption may vary across different visa categories and economic cycles.

Our unemployment results are also related to the preliminary work of Alinaghi and Ball (2025), who show that migration contributes significantly to labour force participation in New Zealand. Specifically, they report that the foreign-born employment rate has increased by 302% since 2000 compared to 11% for natives, while the overall positive impact on employment is 54%.²⁰ They also find that foreign-born residents are more than twice as likely to hold university qualifications than natives, 38.2% versus 17.8%, which might result in an aggregate fall in the unemployment rate in response to an increase in net migration. These results are consistent with our findings that migration is a major driver of employment growth in New Zealand.

²⁰Alinaghi and Ball (2025) urge caution in interpreting these numbers due to data limitations.

The fall in the unemployment rate may be explained by a greater demand for labour in response to immigrants' additional demand for goods and services. This may lead to job creation (Fromentin, 2013), where immigrants create more jobs than they fill due to increased demand (Altonji & Card, 1991; Simon, 1989). The fall in the unemployment rate may also reflect the complementarity between the skill sets of immigrant and native workers (Ottaviano & Peri, 2008; Peri & Sparber, 2009). Our response of the unemployment rate on impact contrasts with some of the findings in the literature. For instance, in the DSGE models of Furlanetto and Robstad (2018) and Kiguchi and Mountford (2019), an immigration shock is modeled as an exogenous increase in population, where immigrants enter as searchers. In this case, some of the immigrants find a job within the quarter whereas some become unemployed.²¹ Moreover, the SVAR results of Kiguchi and Mountford (2019) also find a temporary increase in the unemployment rate for the US in response to an increase in net migration. In their model unemployment increases due to the lagged reaction of job vacancies in reaction to a migration shock.²²

Now we turn our attention to the IRFs of the restricted variables depicted in Figure 2. The response of the immigration rate corresponds to qualitatively similar hump-shaped dynamics of real GDP and the participation rate, while real wages fall in the short-term. One possible explanation is that an increase in the immigration rate generates economic activity resulting in an increase in the participation rate of domestic workers, which translates into a fall in real wages. Quantitatively, we see an expansion in real GDP of around 0.4% six months after the expansionary immigration shock, while it remains significant for around 18 months. Overall, the positive response of GDP to an immigration shock is highly persistent, which may partly be due to a rise in aggregate demand due to increased migrant population. Moreover, as mentioned earlier, migrants seem to actively participate in the labour market and complement native workers in their skill sets, which propagates economic activity.²³

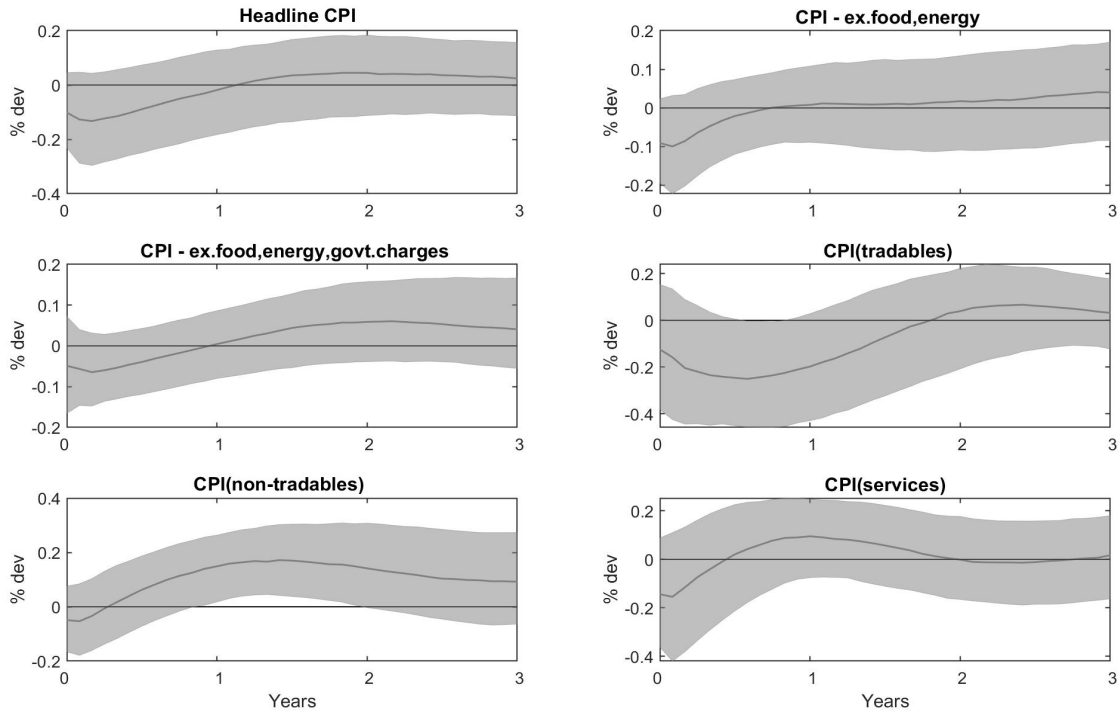
The sign restrictions imposed to identify the immigration shock characterise it as a foreign labour supply shock. As a result, we observe a positive impact response of the participation rate of around 0.10%, and a negative impact response of real wage levels of around 0.18%. For the participation rate, we observe a significantly positive response for more than a year after an expansionary immigration shock, while real wages remain significantly negative for at least the first four months. However, in the long-term, the effect on real wages is likely to be positive, as the major mass of probability intervals lies in positive territory. These findings are corroborated by the recent work

²¹A decline in unemployment can be obtained in the DSGE model if immigrants enter the labour market already with a job (directly into the employment pool) and not as searchers (Furlanetto & Robstad, 2018).

²²Also see Damette and Fromentin (2013) for a related study on a panel of European and Anglo-Saxon countries.

²³Similar findings are reported in Muysken and Zieseimer (2013).

Figure 3: Baseline results: IRFs to a 0.25% immigration shock.



Key insight: Responses of various measures of CPI in levels are statistically insignificant in response to an expansionary immigration shock, except for the significant increase in CPI(non-tradables) after 1 year.

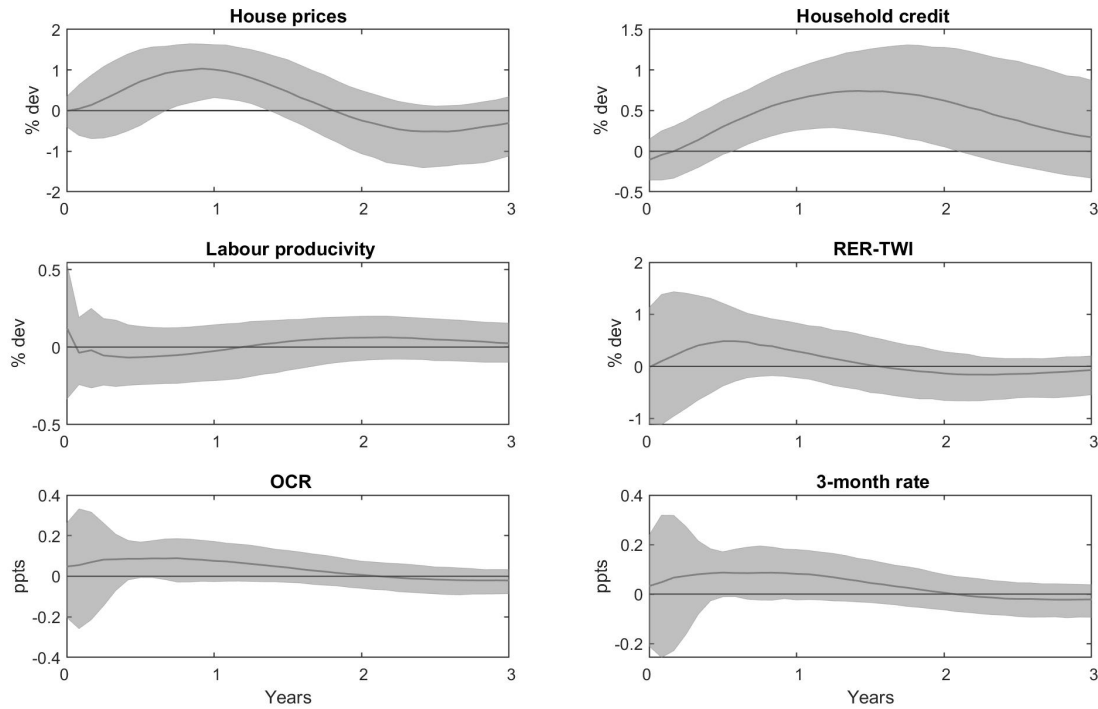
of Alinaghi (2025), who shows that immigrants face an initial wage penalty of approximately 40% relative to native-born workers, but this gap closes almost entirely within 10 years.²⁴

Our next set of results in Figure 3 focuses on various measures of price levels. We replace the unemployment rate with each measure of the consumer price levels separately and estimate the models one at a time. In this way our model remains fully identified while leaving the responses of our variables of interest unrestricted. IRFs of the restricted variables are qualitatively and quantitatively very similar to what we see in Figure 2.²⁵ From Figure 3, headline consumer price levels show no significant deviation from baseline in response to an expansionary immigration shock. The sub-component analysis of consumer prices shows that non-tradable prices are significantly elevated (at the 68% level) approximately one year after the shock. Tradable prices, by contrast, are likely to be below the baseline, with most of the posterior probability mass lying in negative

²⁴For the first time in New Zealand's modern immigration history, average immigrant wages crossed above native wages. By 2023, immigrants earn approximately \$5,100 per month compared to \$5,000 for natives (Alinaghi & Ball, 2025).

²⁵Complete set of IRFs for each model are reported in Appendix B.

Figure 4: Baseline results: IRFs to a 0.25% immigration shock.



Key insight: A statistically significant increase in the house price levels after 2 quarters in response to an expansionary immigration shock, while the median response of OCR is positive but insignificant.

territory. One explanation, based on consumer preferences, may come from immigrants substituting non-tradables for tradables, generating higher demand for non-tradables, which is associated with higher non-tradable price levels. Another possible explanation is the likely impact of rising house prices given that rent and home-ownership have approximately 40% weight in measuring non-tradables price index. Moreover, the responses of the sub-components of price levels show that in the short-term, the headline consumer price levels response is largely similar to the responses of tradables and services. However, in the medium-term, tradable prices are likely to fall, which is offset by the statistically significant increase in the prices of non-tradables. As a consequence, the response of headline consumer price levels remains insignificant throughout the horizon.

Figure 4 reports the results for other variables of interest, again substituting the unemployment rate for each of the remaining variables separately. The most important results are significant increases in the levels of house prices and household credit, which unsurprisingly correspond with the hump-shaped responses of immigration and economic activity. The responses of the OCR and the 3-month interbank rate show that the central bank is likely to increase the policy rate in the

medium-term, which is associated with lower price levels - a partial explanation of a fall in the headline consumer price levels in the short to medium-term as reported in Figure 3. Further, Figure 4 shows a likely appreciation of the domestic currency. This may improve the terms of trade through lower import prices, exerting downward pressure on consumer price levels, particularly for tradables.²⁶ Although we do not consider terms of trade, export prices, import prices and private consumption in our model, given that it is not the focus of this study, it appears that the exchange rate channel may play a role in the transmission of immigration shocks.²⁷

Now we turn to the response of labour productivity. To perform this analysis, we replace the unemployment rate with labour productivity, defined as the ratio of real GDP to total hours worked per week. Since New Zealand is characterised by high-skilled immigration, one would expect immigration to have a positive impact on labour productivity and consequently a positive impact on long-term economic growth. However, the IRF depicted in Figure 4 reports otherwise: a muted response of labour productivity to an expansionary immigration shock. This result is in line with what the New Zealand Productivity Commission concluded in its 2022 report: “immigration is likely to be neither the solution to, nor the cause of, the productivity challenges facing New Zealand” (New Zealand Productivity Commission, 2022).²⁸ Moreover, Alinaghi and Ball (2025) report that the recent composition of immigrants has shifted toward lower-skilled temporary workers, particularly in the agriculture and hospitality sectors, which could have a negative long-term productivity implication. On the other hand, given migrants’ skill composition and their degree of complementarity to that of native workers, immigration may also make the fall in labour productivity less pronounced. We shed more light on labour productivity in our robustness analysis in Section 5, where we investigate the impact of two immigrant age groups on the New Zealand economy.

4.2 Forecast Error Variance Decomposition (FEVD)

Next, we report the FEVD analysis, which shows how much of the observed variation is explained by the immigration shock. These decompositions are depicted in Figure 5 for the baseline model using the unemployment rate.²⁹ The immigration shock contributes the most to the immigration rate, explaining more than 40% of the observed variation. The immigration shock explains between 10% and 30% of the variation in real GDP levels and the unemployment rate over a three year horizon. For the participation rate, the contribution of the immigration shock increases sharply, reaching 25% after three years. Considering the FEVD of real wage levels, the proportion of variation ex-

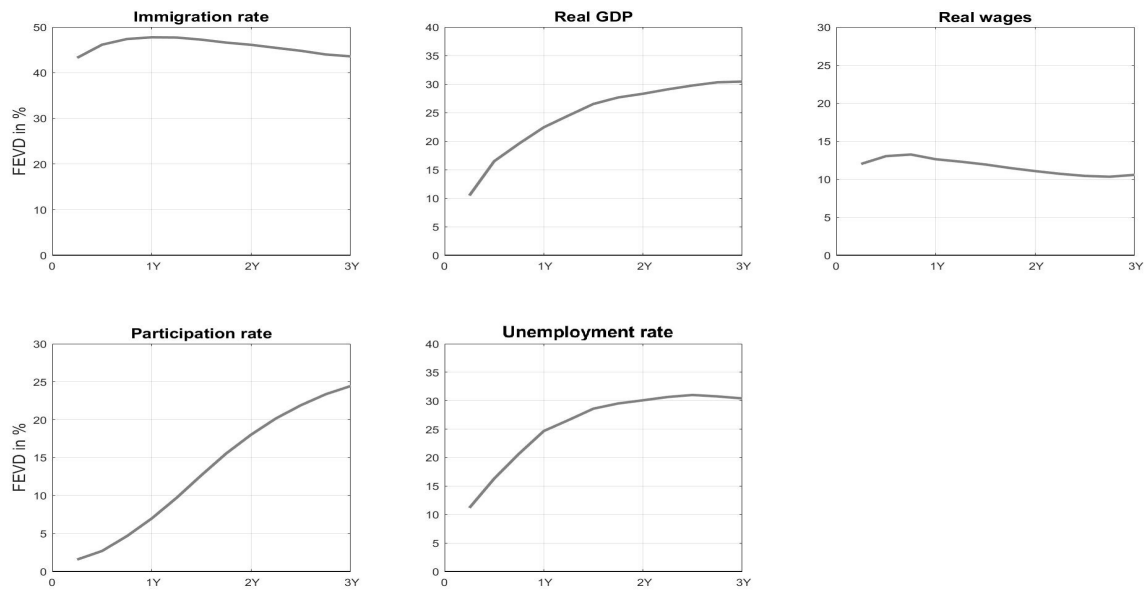
²⁶We define a rise in the exchange rate as an appreciation of the domestic currency.

²⁷See Smith and Thoenissen (2019).

²⁸Also see New Zealand Productivity Commission (2021) report.

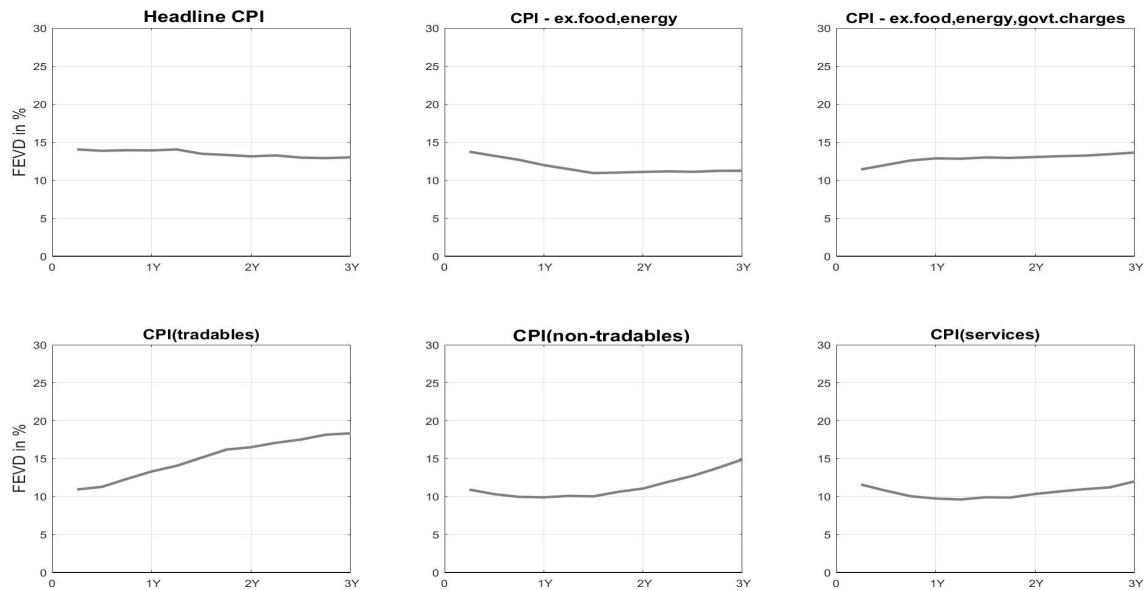
²⁹The full set of FEVD results are in Appendix C.

Figure 5: Baseline results: FEVD to the immigration shock.



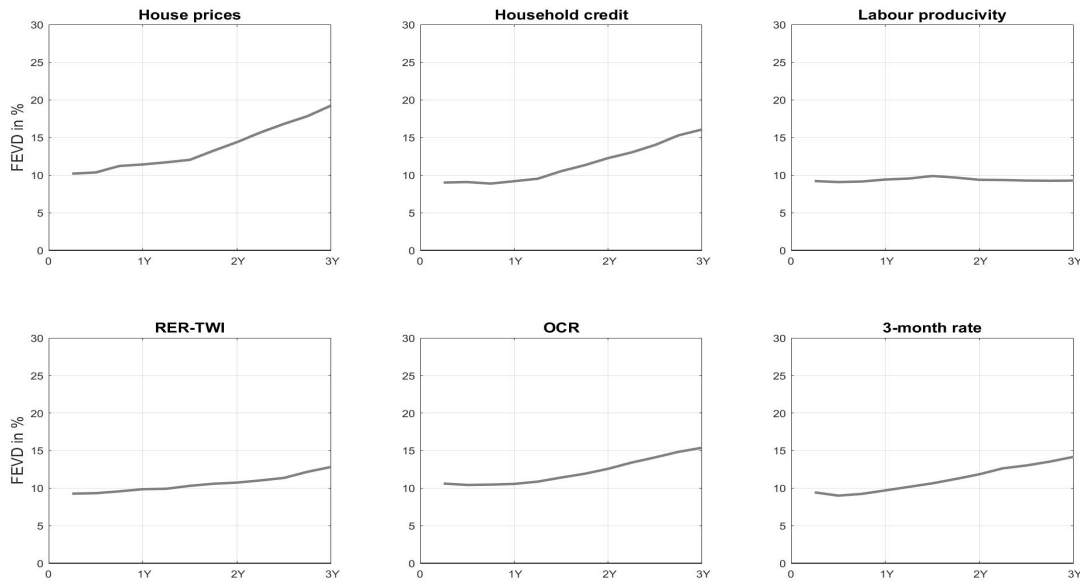
Key insight: The immigration shock is a major contributor of 30% of the variation in real GDP and unemployment rate at the longer horizon.

Figure 6: Baseline results: FEVD to the immigration shock.



Key insight: The immigration shock contributes negligibly towards the variations in headline consumer price levels throughout the horizon.

Figure 7: Baseline results: FEVD to the immigration shock.



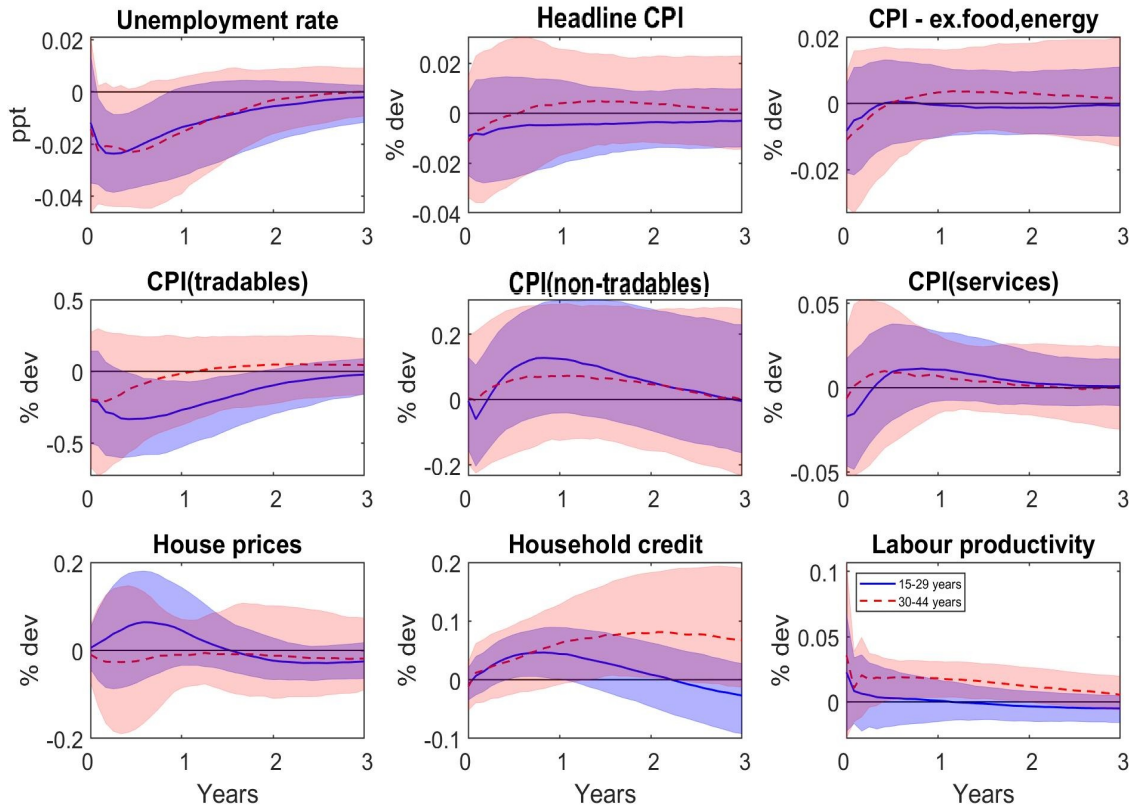
Key insight: The immigration shock is a non-negligible contributor towards the variations in house price levels at longer horizon.

plained by the immigration shock remains between 10% and 15%. From Figure 6, the contribution of the immigration shock to the headline consumer price levels (10%), tradable prices (18%) and non-tradable prices (15%) is negligible after three years. From Figure 7, the immigration shock explains less than 10% of the variation in labour productivity over the full horizon period, which is less than the proportion of variation explained for the other variables in the model. Immigration shock explains 20% of the variation in house price levels observed at the three year horizon. However, the contribution of the immigration shock towards the household credit level (16%), the policy rate (15%), and the exchange rate (13%) remains negligible.

5 Robustness check

We perform a robustness check to evaluate the impact of expansionary immigration shocks, where the immigration rate is computed using two age groups: “younger” classified as 15-29 years and “older” classified as 30-44 years.

Figure 8: Robustness check: IRFs to a 0.25% immigration shock.



Key insight: The unemployment rate falls significantly in response to an immigration shock to “younger” age group, while labour productivity is likely to increase in case of an immigration shock to “older” age group. *Note:* The solid (dashed) lines are the point-wise posterior median responses for age 15-29 (30-44). The shaded areas represent the respective 68% equal-tailed posterior probability bands.

5.1 Age composition of immigrants: 15-29 and 30-44 years

In this analysis, we estimate two SVAR models based on the age of immigrants.³⁰ Specifically, we study the responses of our aggregate macroeconomic variables to “younger” and “older” immigration shocks, where we classify “younger” migrants as those aged 15-29 and “older” migrants as those aged 30-44. The IRFs are displayed in Figure 8. The solid (dashed) lines are point-wise posterior median IRFs in response to a younger (older) immigration shock, while the shaded regions are the respective 68% equal-tailed posterior probability bands. In response to a “younger” immigrant shock, we observe a statistically significant fall (using the 68% credible intervals) in the unemploy-

³⁰We continue to use the baseline specification for this analysis.

ment rate and the level of tradable prices, while house prices are likely to increase given a major mass of probability intervals lie in the positive region. In response to an “older” immigration shock, the IRFs of all variables are largely insignificant except for labour productivity. The median response of labour productivity remains positive throughout the horizon and shows a statistically significant increase (at the 68% credible interval) at around the 1 year horizon. Moreover, a large mass of the probability intervals of labour productivity lie in the positive territory, which implies that labour productivity is likely to increase following an expansionary immigration shock for immigrants aged 30-44 years.³¹

6 Conclusion

We study the impact of an expansionary immigration shock on the New Zealand labour market, various measures of consumer price levels, house price levels, and labour productivity using sign-restricted SVAR models. Our findings indicate that the immigration rate, real GDP levels, and the participation rate initially increase significantly and then subsequently decrease in response to an expansionary immigration shock. Real wage levels fall temporarily, while the decrease in the unemployment rate is persistent and significant.

Moreover, we perform a number of exercises which show that expansionary migration shocks do not have a significant impact on various measures of consumer price levels, except for a statistically significant increase in non-tradable prices (at 68% credible intervals) in the medium-term. Further analysis shows a statistically significant increase in the levels of house prices and household credit, while, surprisingly, we find a statistically insignificant response of labour productivity even though New Zealand receives highly skilled immigrants. Our FEVD analysis shows that immigration shocks contribute mainly towards fluctuations in real GDP levels, as well as the participation rate and the unemployment rate.

Our robustness analysis highlights two key findings. First, the fall in the aggregate unemployment rate is more pronounced when relatively younger immigrants enter the labour market. Second, labour productivity is likely to increase with an increase in the inflow of relatively older immigrants.

Several limitations should be considered when interpreting these results. First, the assumption of rapid labour market integration may not apply uniformly across all migrant categories, particularly those on family reunification or student visas. Second, the relatively short sample period may

³¹The FEVD results in Appendix D show that the overall contribution of an immigration shock towards the variables of interest for both age groups remains negligible (less than 20%).

reduce the statistical power of our estimates, potentially hindering the detection of smaller effects. Third, recent shifts in migration dynamics suggest that these results may not fully generalise to the post-COVID-19 era. Fourth, the absence of granular data on skill levels and prior experience obscures the relationship between age and expertise, leaving it unclear whether productivity improvements are driven by the age of the migrants or their professional background.

However, these constraints provide a framework for future research. One promising avenue involves using micro-level data on age, education, and expertise to determine if higher human capital enhances productivity independently of age. Additionally, researchers could extend the sample period to include the COVID years by employing recent Bayesian SVAR techniques, such as the 'Pandemic Priors' developed by Cascaldi-Garcia (2025). Such methods would strengthen the statistical power of the estimates and offer deeper insights into the New Zealand's macroeconomy responses to evolving migration patterns. Another interesting avenue would be to explore the exchange rate channel, investigating the change in consumer consumption patterns associated with changes in export prices, import prices, and the terms of trade in response to an increase in net migration. This would provide insight into the demand-side effects on tradable and non-tradable goods, as well as the response of monetary policy.

Finally, our research has two key policy implications. First, we observe that the unemployment rate falls and real GDP levels increase in response to an expansionary immigration shock. This reflects increased macroeconomic activity. Second, an expansionary immigration shock increases house price levels and household credit, reflecting another channel that generates economic activity. In sum, expansionary immigration shocks have been associated with economic growth, reduced unemployment, and increased housing market activity in New Zealand over our sample period.

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Appendix: "Migration and the New Zealand economy: Evidence from set-identified SVARs."

Naveed Javed, India Power and Adam Richardson

A Data Sources

The dataset spans from 1992 $q1$ to 2019 $q4$. Data is last collected in August 2024.³²

- **Immigration rate** - Ratio of stock of immigrants to working age population (WAP). The stock of immigrants is accumulated net migration. We concatenate the two series: net quarterly PLT migration (data from 1982-2018) and net estimated migration (data from 2018 - 2019). In order to avoid negative entries in the stock of immigrants, we scale it by the corresponding starting value of annual permanent and long-term migration key series from Stats NZ.³³
- **Production-based gross domestic product (GDP)** (Real, Seasonally Adjusted) - QGDP Gross Domestic Product - Production measure with RBNZ estimate before Jun-87. *Source:* RBNZ
- **Real wages** (Real, Seasonally Adjusted) - QES: Average hourly earnings (ordinary time): Private Sector. Nominal wages are deflated using the core CPI measure from FRED. *Source:* Stats NZ.
- **Participation rate** (Seasonally Adjusted). *Source:* Stats NZ.
- **Unemployment rate** (Seasonally Adjusted). *Source:* Stats NZ.
- **Consumer Price Index - All groups** (Seasonally Adjusted). *Source:* Stats NZ.
- **Sub-components of Consumer Price Index - tradeables, non-tradables, services** (Seasonally Adjusted). *Source:* Stats NZ.
- **Core Consumer Price Index** (Seasonally Adjusted). All groups less food group, household energy subgroup, vehicle fuels. *Source:* FRED Economic Data (CPGRLE01NZQ657N).
- **Core Consumer Price Index (1999 $q2$ -2019 $q4$)** (Seasonally Adjusted). All groups less food group, household energy subgroup, vehicle fuels, & government charges. *Source:* Stats NZ.
- **House prices** (Seasonally Adjusted). *Source:* Real Estate Institute of New Zealand (REINZ).
- **Household credit** (Nominal, Seasonally Adjusted). Banks & NBLIs: Housing & personal consumer lending - Total (break adjusted). Household credit deflated by the core CPI measure from FRED. *Source:* RBNZ - Bank balance sheet.

³²Series are seasonally adjusted using Eviews Census X-13

³³We do not test for the structural break in 2018 assuming that there will be insignificant difference in estimation from the four observations of 2019.

- **Labour productivity** (Seasonally Adjusted). Ratio of real GDP to total hours worked per week. The total number of active hours worked each week scaled by 13 to make it consistent with the quarterly GDP. *Source:* Stats NZ.
- **Official cash rate** (Percent, Seasonally Adjusted). *Source:* RBNZ.
- **3-Month or 90-day Rates and Yields: Interbank Rates for New Zealand** (Percent, Not Seasonally Adjusted). *Source:* FRED Economic Data (IR3TIB01NZQ156N).
- **Exchange rate** *Source:* RBNZ.
- **Working age population (WAP)**. Population aged between 15 and 65. *Source:* Stats NZ.

B Baseline - IRFs

B.1 IRFs of the immigration shock using Headline CPI

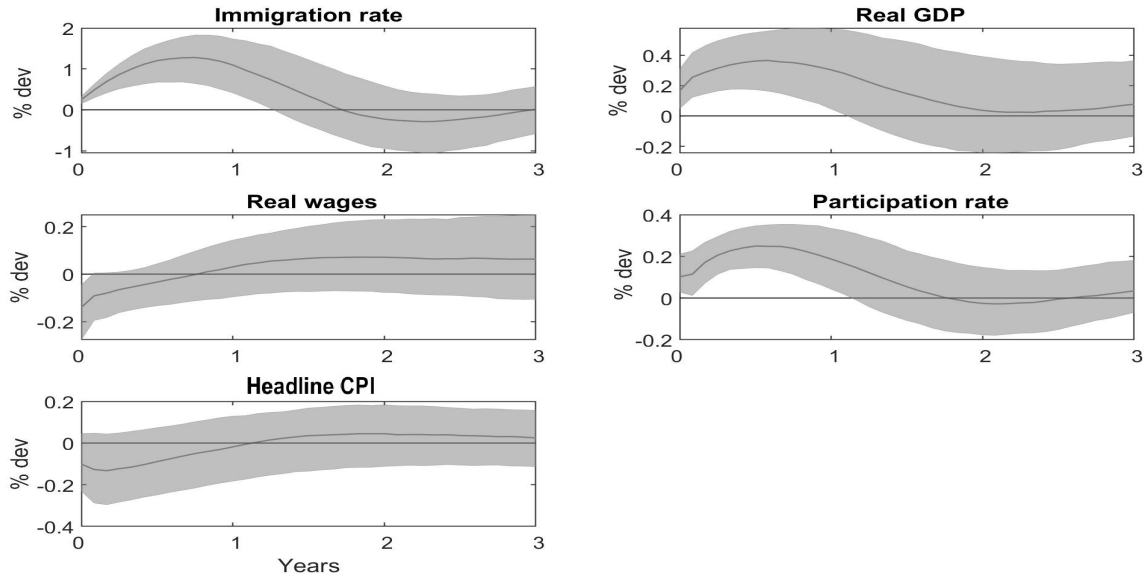


Figure 1: IRFs to a 0.25% expansionary immigration shock using Headline CPI

B.2 IRFs of the immigration shock using Core CPI (FRED measure)

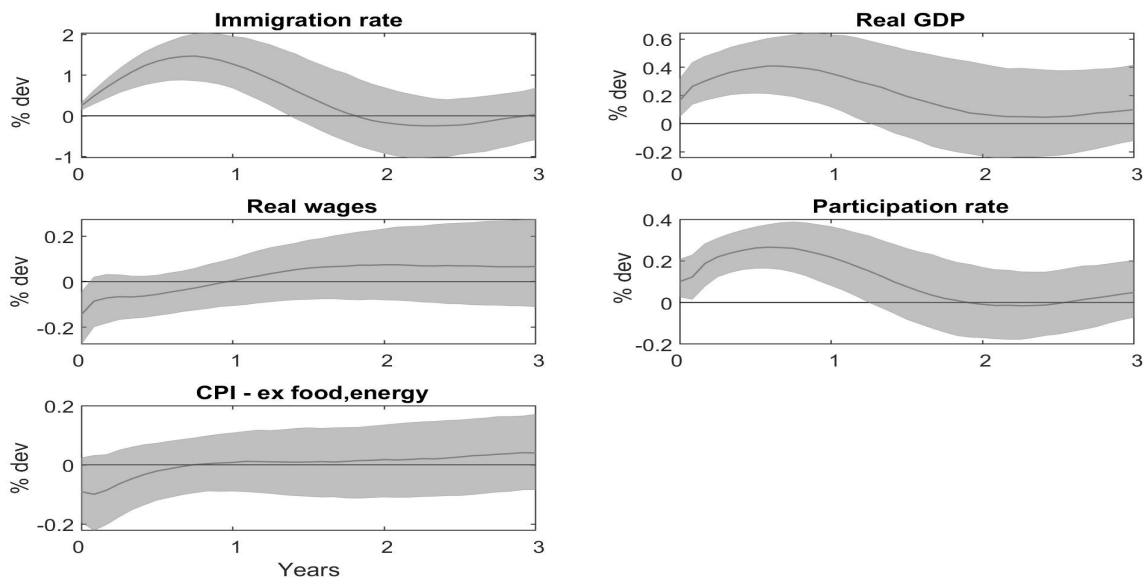


Figure 2: IRFs to a 0.25% expansionary immigration shock using Core CPI (FRED measure)

B.3 IRFs of the immigration shock using Core CPI (Stats NZ): 1999q2-2019q4

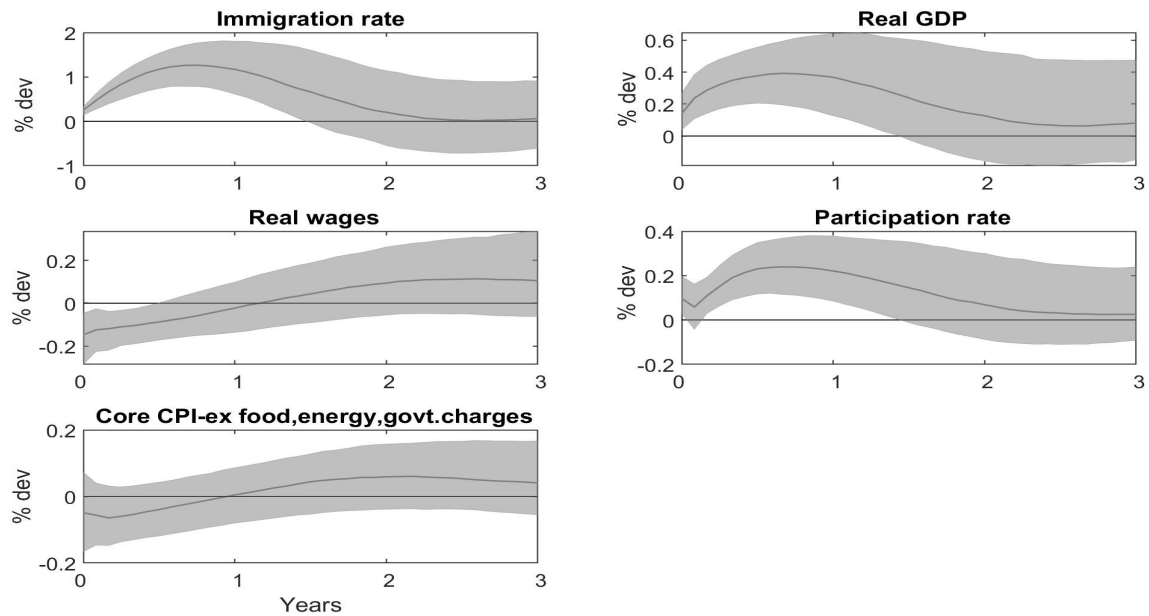


Figure 3: IRFs to a 0.25% expansionary immigration shock using Core CPI (Stats NZ)

B.4 IRFs of the immigration shock using CPI (tradables)

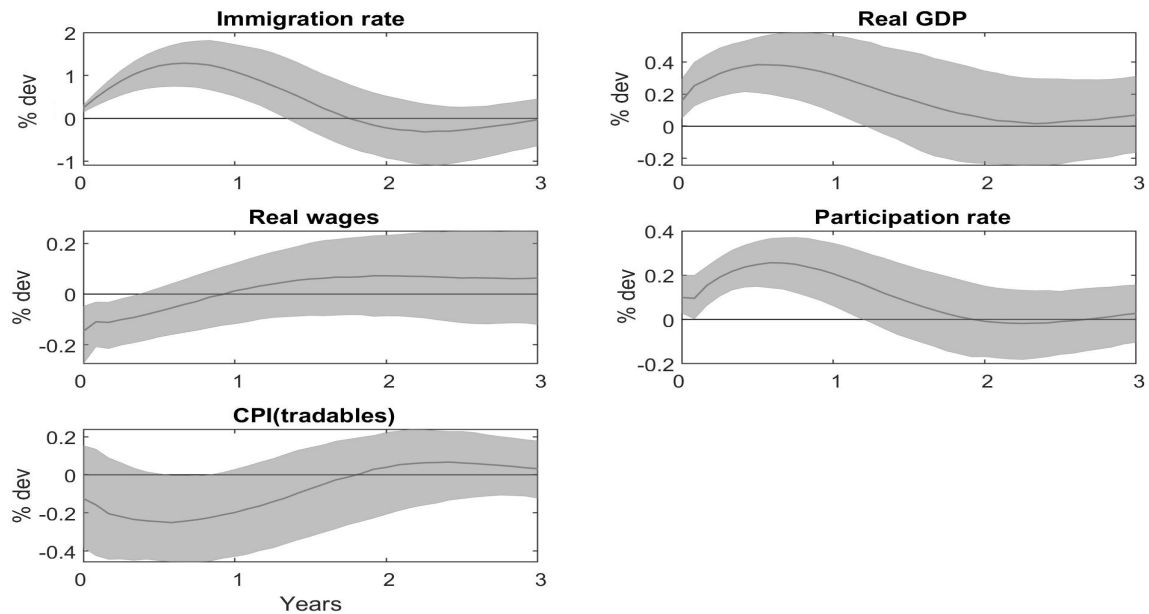


Figure 4: IRFs to a 0.25% expansionary immigration shock using CPI (tradables)

B.5 IRFs of the immigration shock using CPI (non-tradables)

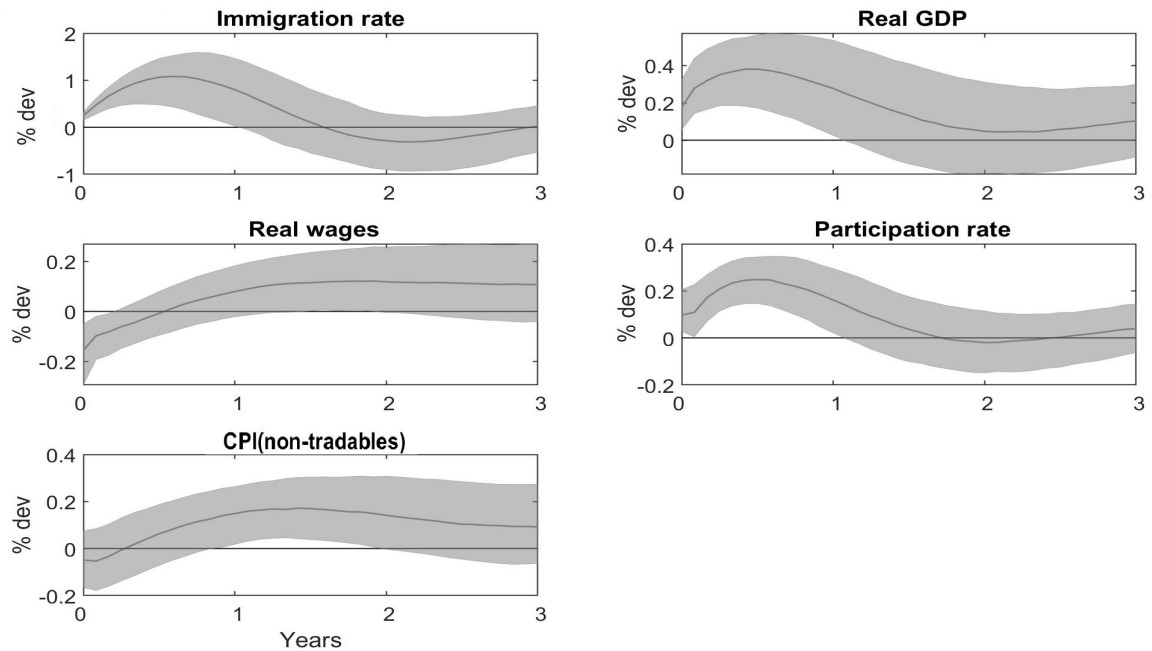


Figure 5: IRFs to a 0.25% expansionary immigration shock using CPI (non-tradables)

B.6 IRFs of the immigration shock using CPI (services)

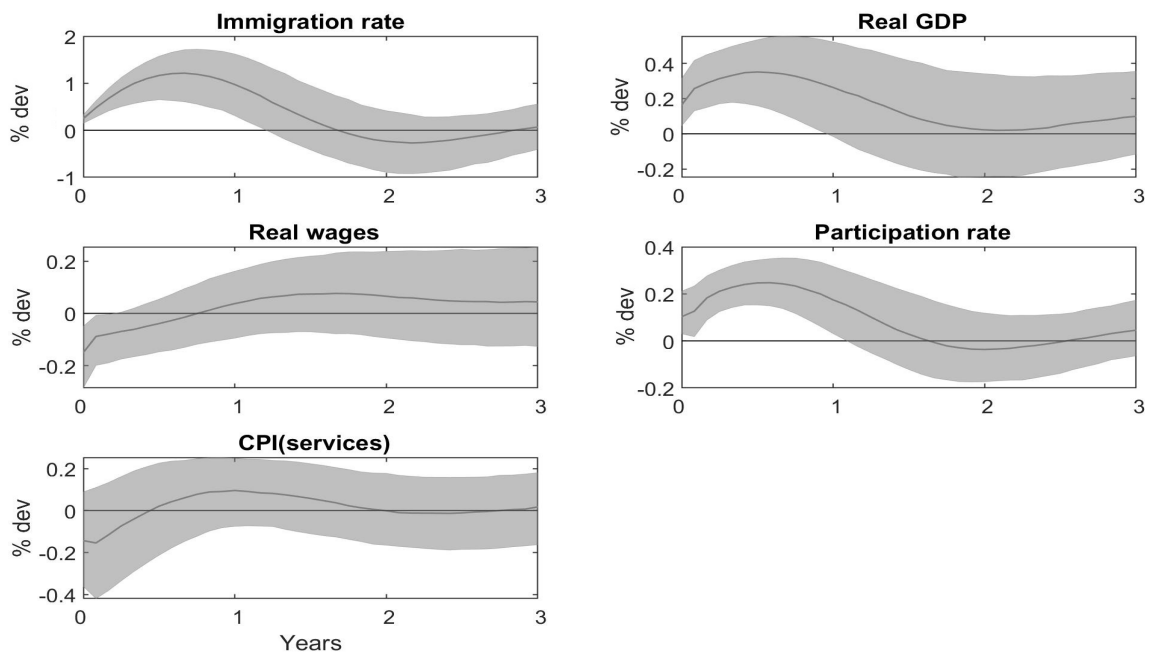


Figure 6: IRFs to a 0.25% expansionary immigration shock using CPI (services)

B.7 IRFs of the immigration shock using house prices

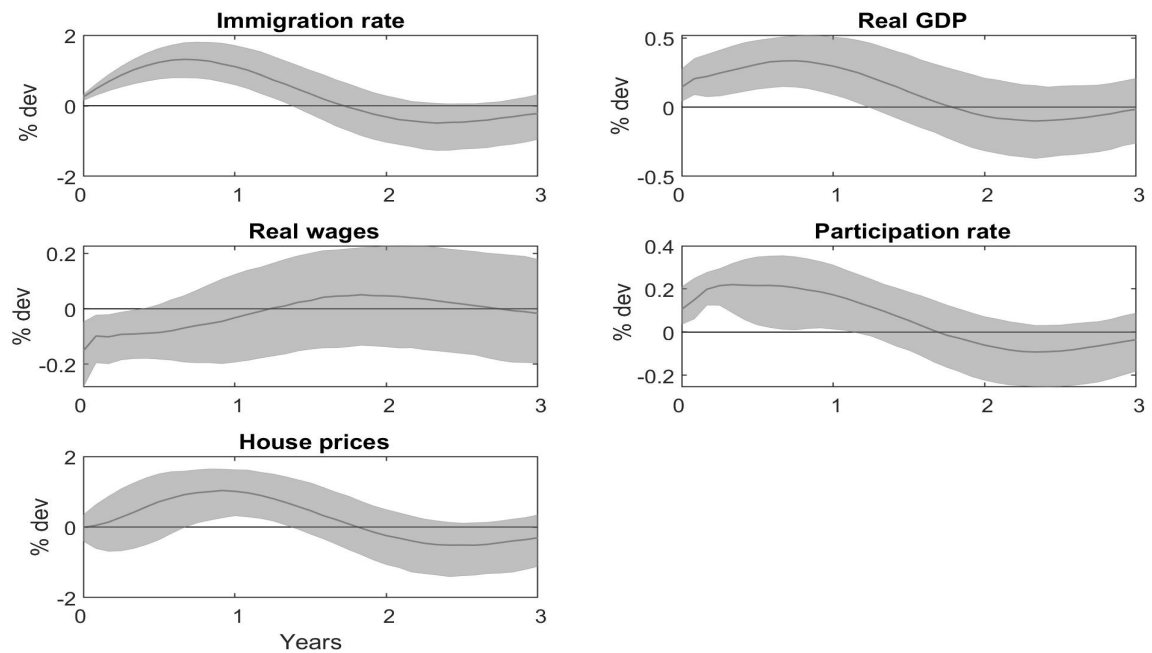


Figure 7: IRFs to a 0.25% expansionary immigration shock using house prices

B.8 IRFs of the immigration shock using household credit

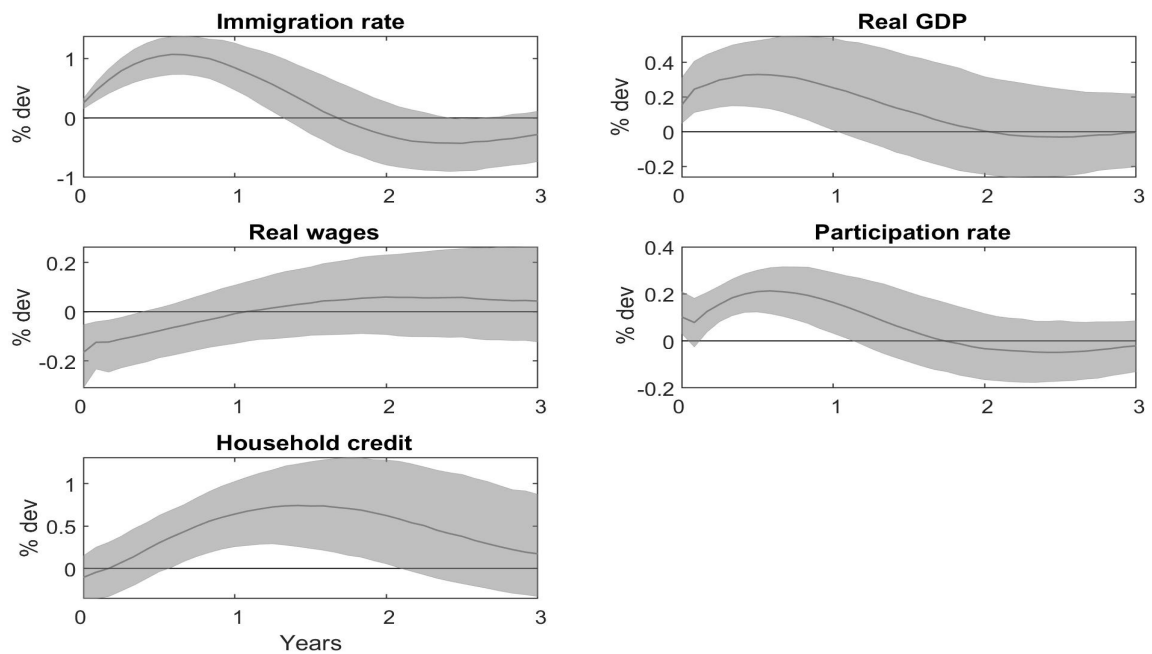


Figure 8: IRFs to a 0.25% expansionary immigration shock using household credit

B.9 IRFs of the immigration shock using labour productivity

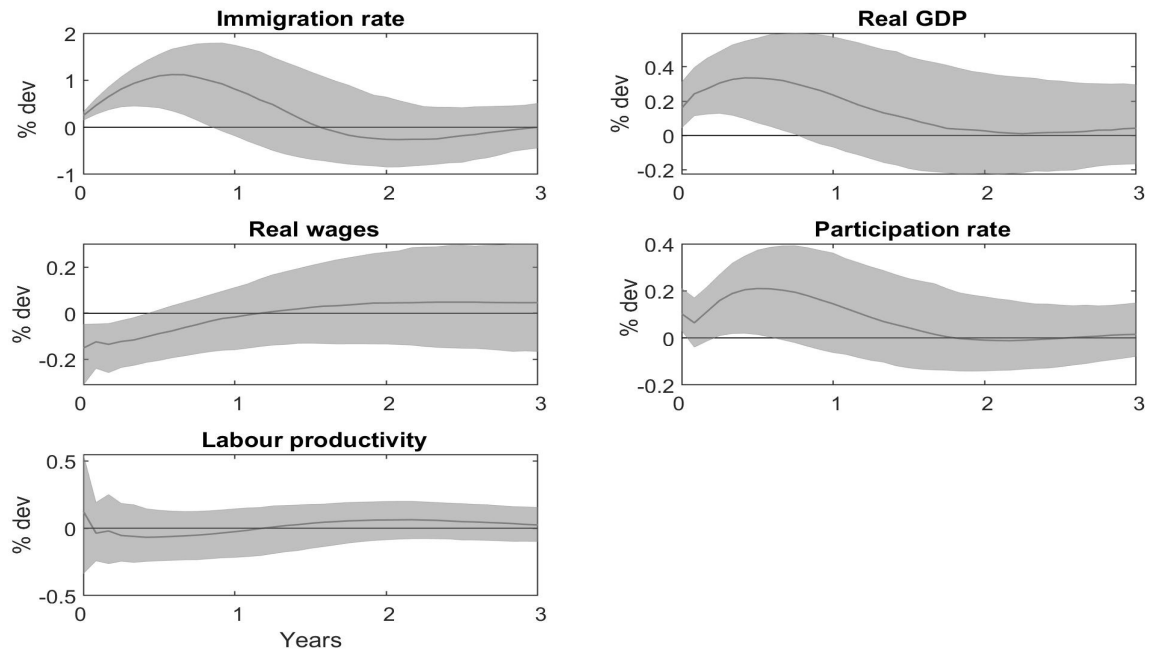


Figure 9: IRFs to a 0.25% expansionary immigration shock using labour productivity

B.10 IRFs of the immigration shock using RER-TWI

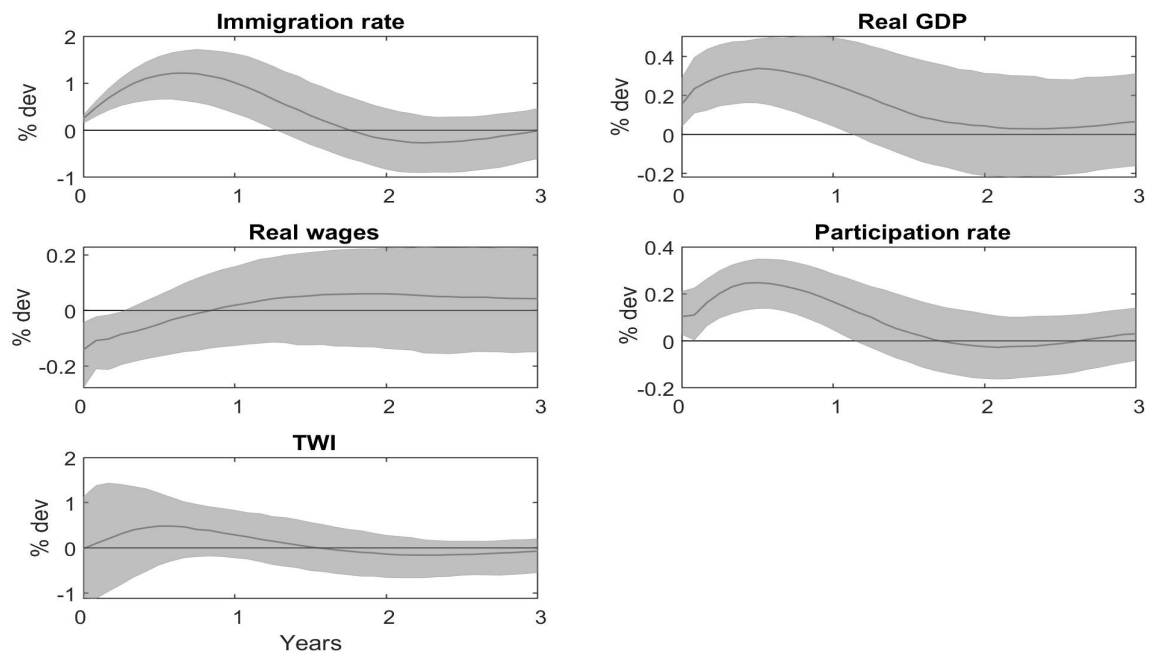


Figure 10: IRFs to a 0.25% expansionary immigration shock using RER-TWI

B.11 IRFs of the immigration shock using OCR

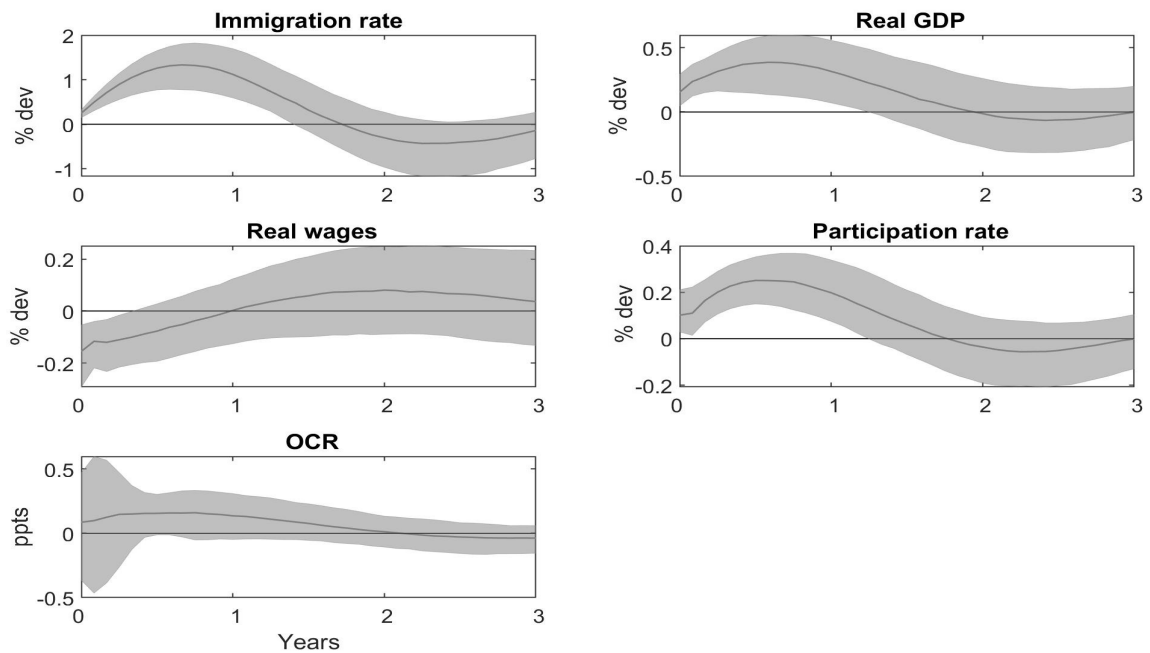


Figure 11: IRFs to a 0.25% expansionary immigration shock using OCR

B.12 IRFs of the immigration shock using 3-Month Interbank Rate

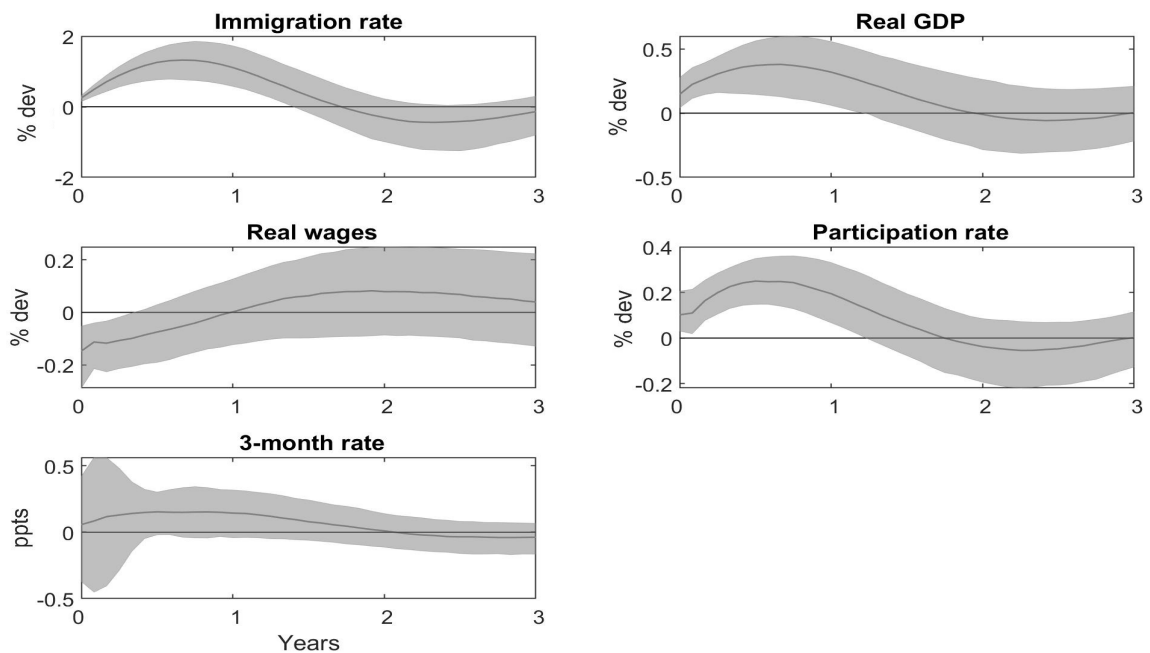


Figure 12: IRFs to a 0.25% expansionary immigration shock using 3-Month Interbank Rate

B.13 IRFs of a domestic labour supply shock using the unemployment rate

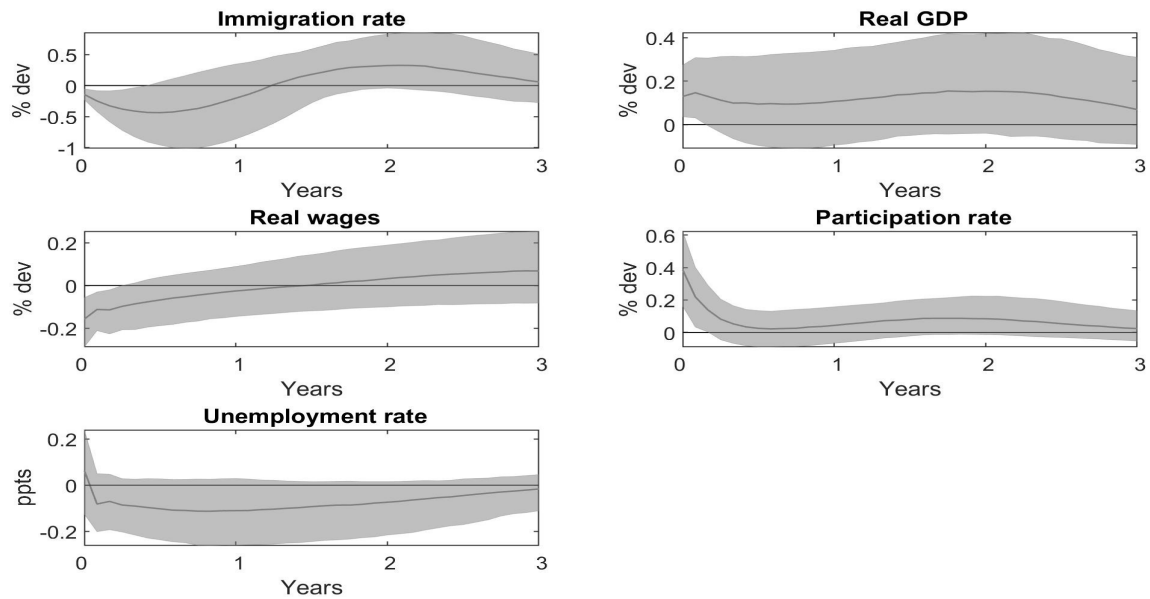


Figure 13: IRFs to a 0.25% domestic labour supply shock using unemployment rate

B.14 IRFs of a domestic labour supply shock using labour productivity

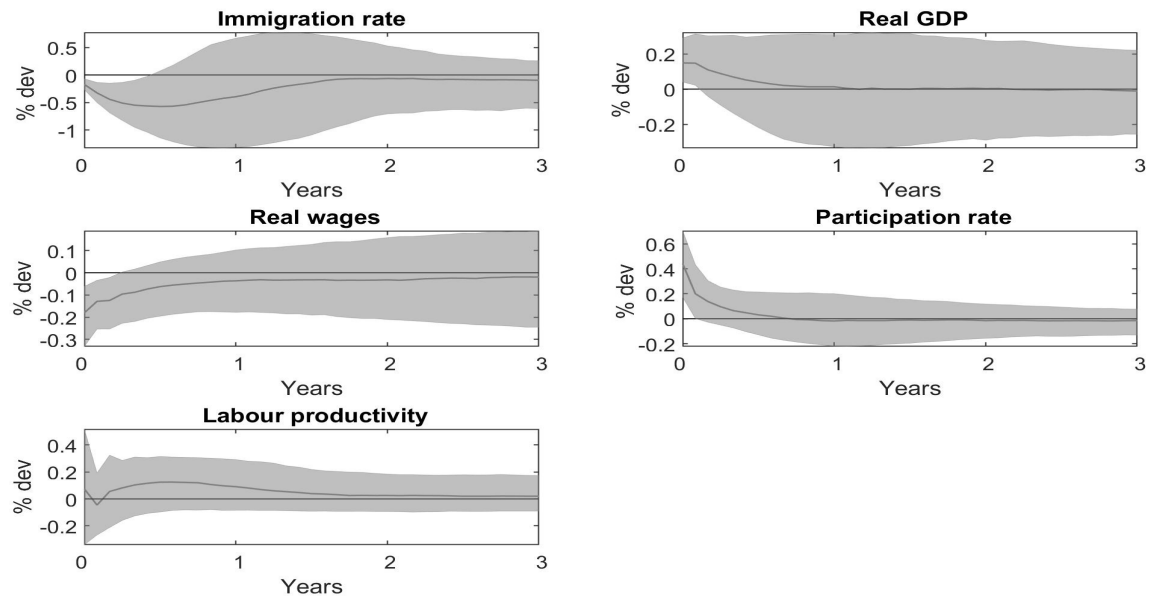


Figure 14: IRFs to a 0.25% domestic labour supply shock using labour productivity

C Baseline - FEVD

C.1 FEVD of the immigration shock using Headline CPI

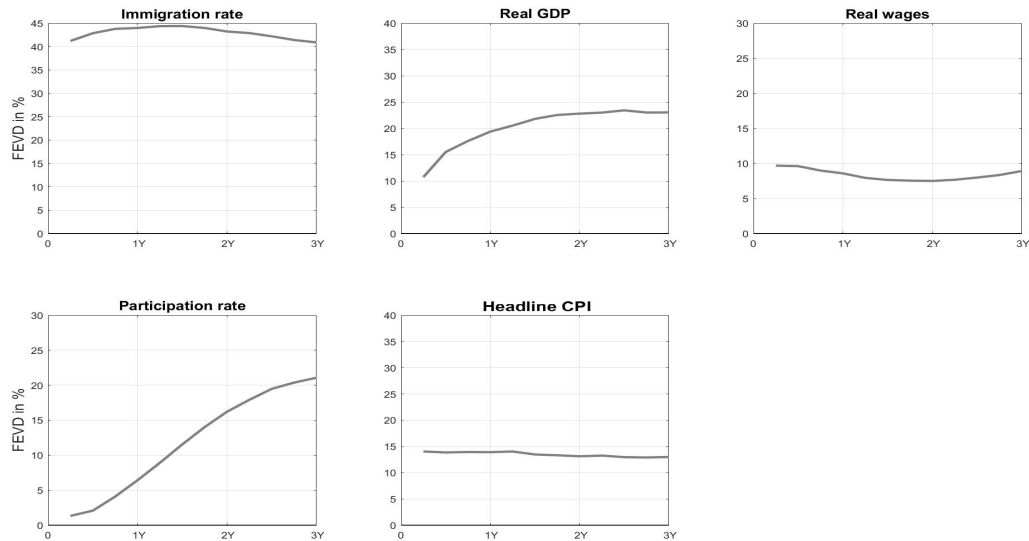


Figure 15: FEVD of the immigration shock using Headline CPI

C.2 FEVD of the immigration shock using Core CPI (FRED measure)

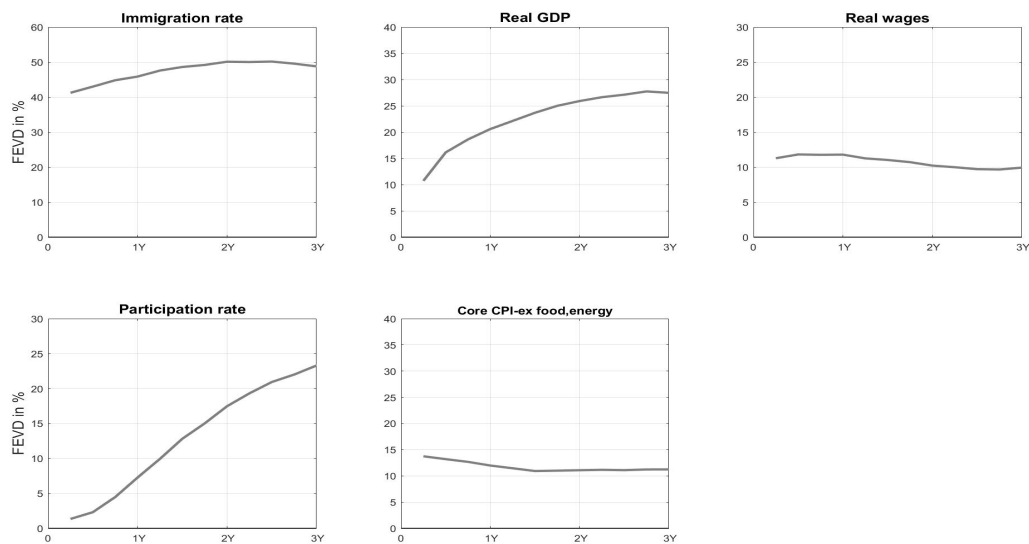


Figure 16: FEVD of the immigration shock using Core CPI (FRED measure)

C.3 FEVD of the immigration shock using Core CPI (Stats NZ): 1999q2-2019q4

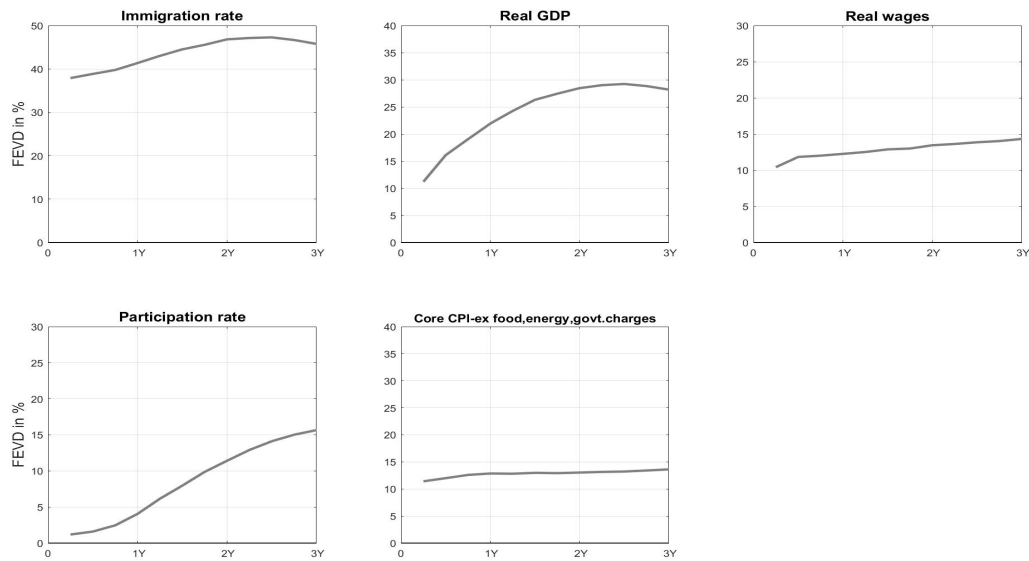


Figure 17: FEVD of the immigration shock using Core CPI (Stats NZ)

C.4 FEVD of the immigration shock using CPI (tradables)

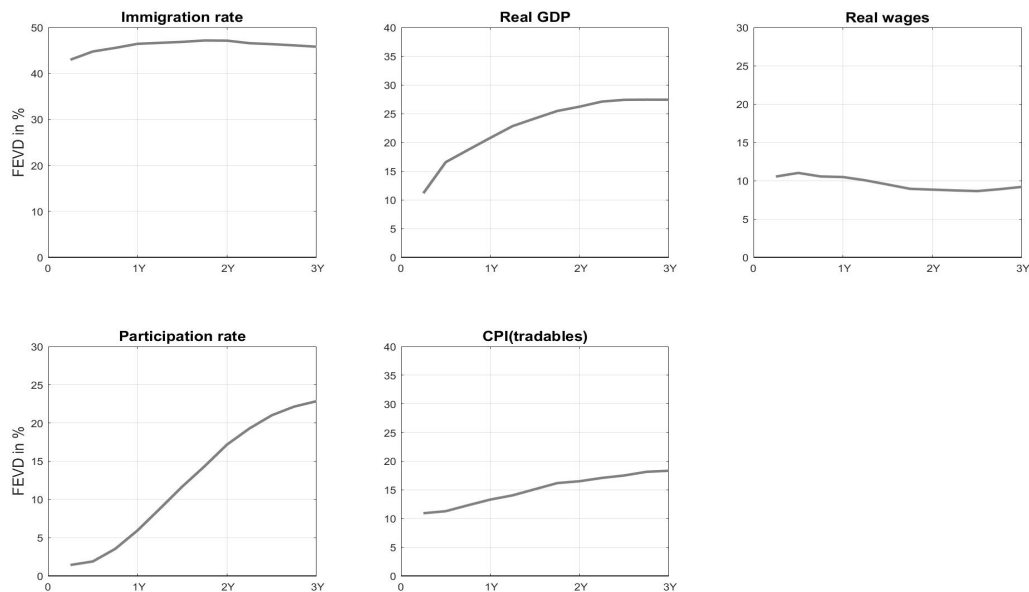


Figure 18: FEVD of the immigration shock using CPI (tradables)

C.5 FEVD of the immigration shock using CPI (non-tradables)

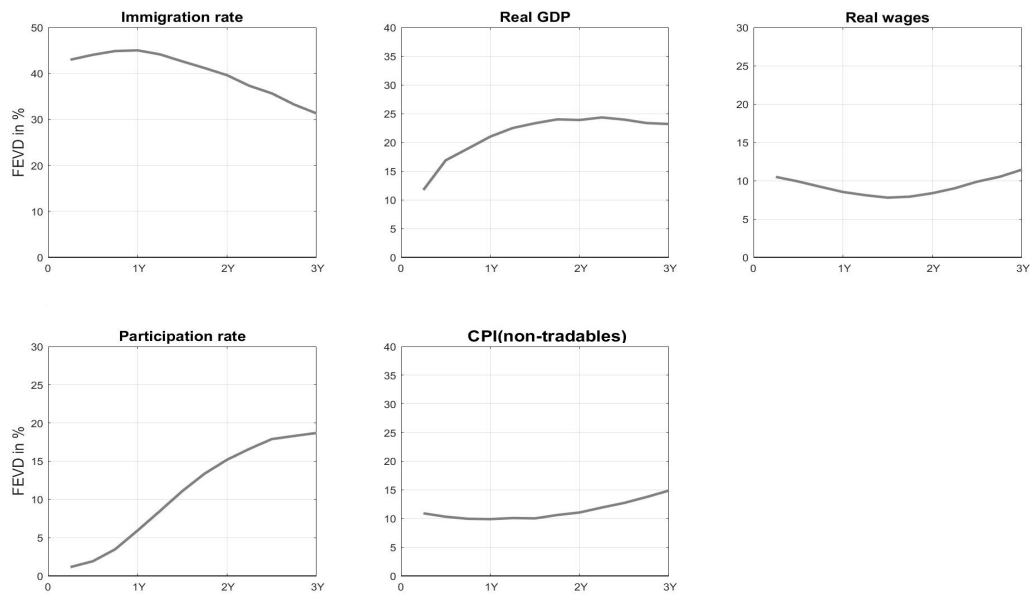


Figure 19: FEVD of the immigration shock using CPI (non-tradables)

C.6 FEVD of the immigration shock using CPI (services)

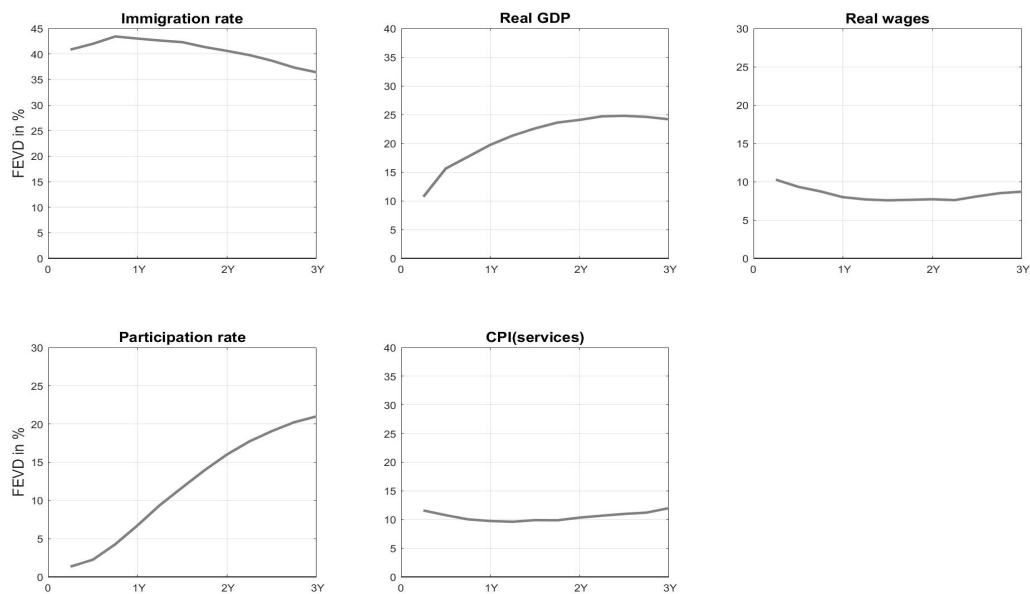


Figure 20: FEVD of the immigration shock using CPI (services)

C.7 FEVD of the immigration shock using house prices

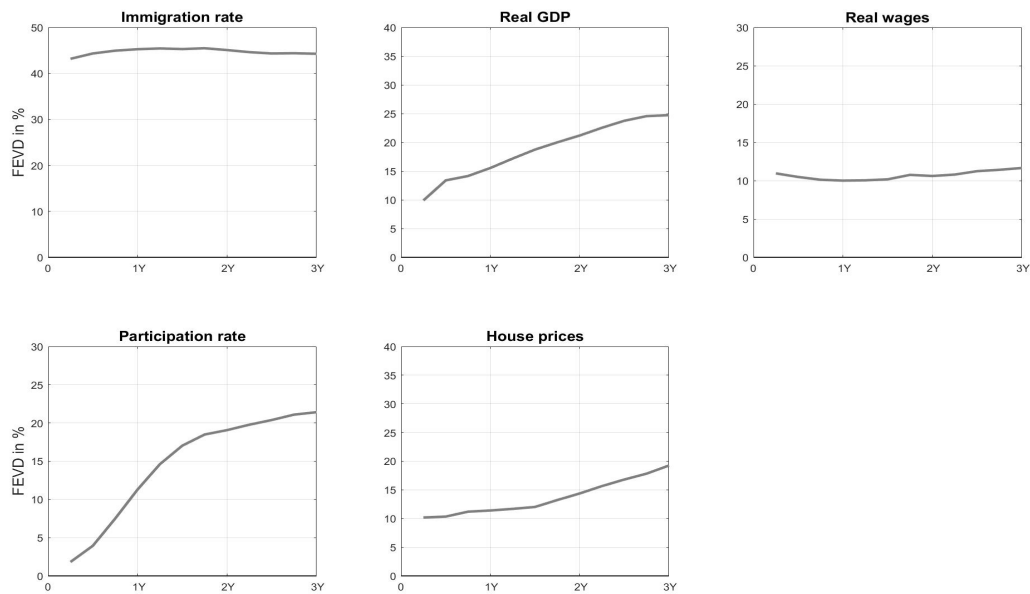


Figure 21: FEVD of the immigration shock using house prices

C.8 FEVD of the immigration shock using household credit

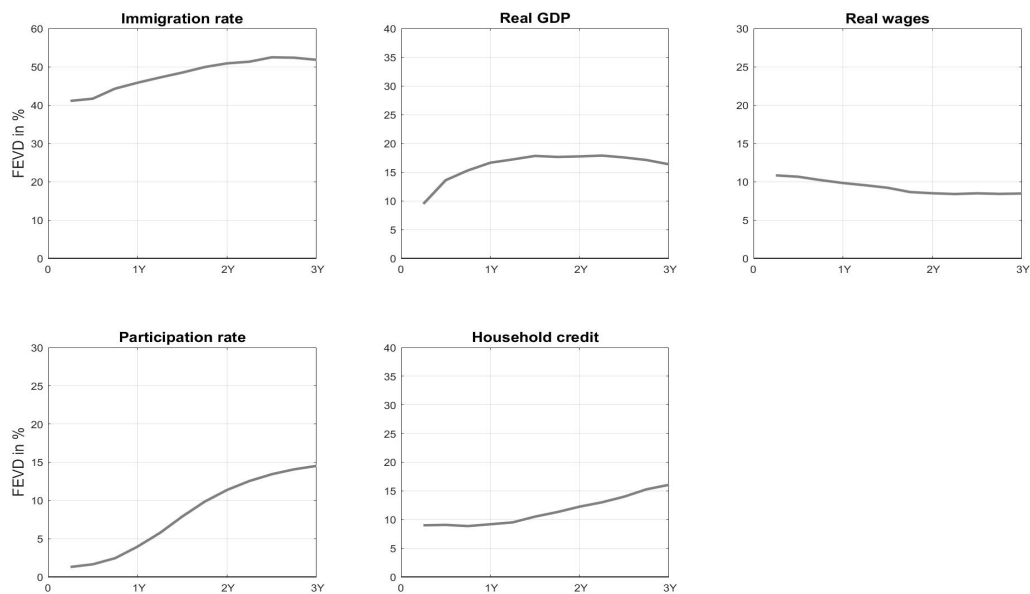


Figure 22: FEVD of the immigration shock using household credit

C.9 FEVD of the immigration shock using labour productivity

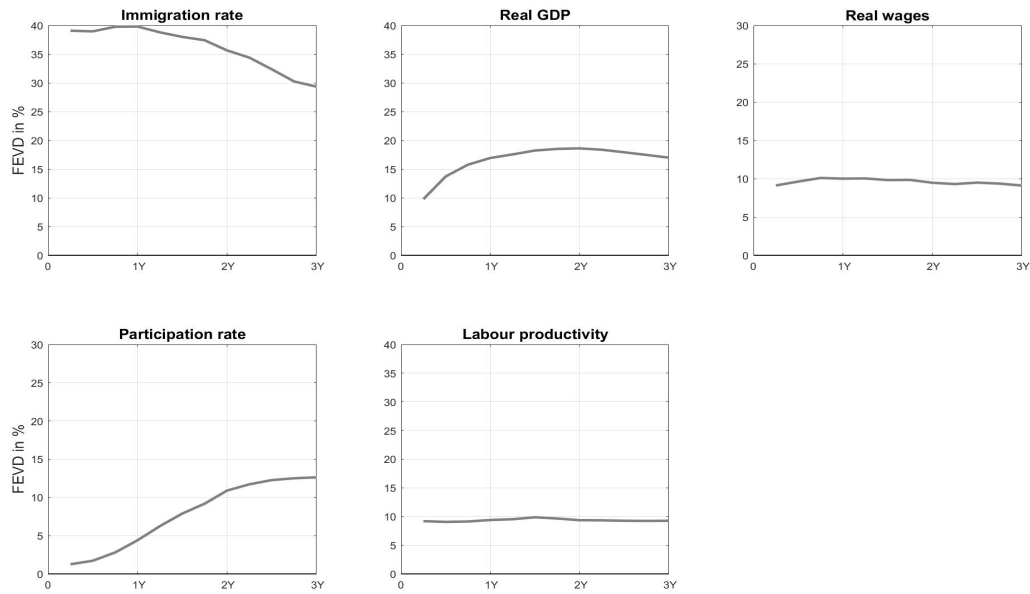


Figure 23: FEVD of the immigration shock using labour productivity

C.10 FEVD of the immigration shock using RER-TWI

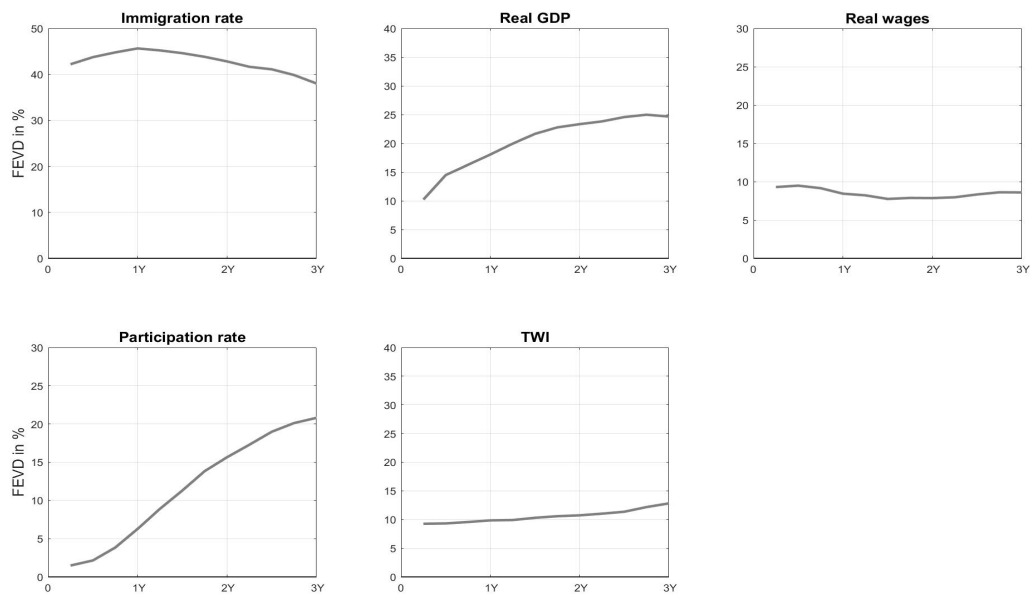


Figure 24: FEVD of the immigration shock using RER-TWI

C.11 FEVD of the immigration shock using OCR

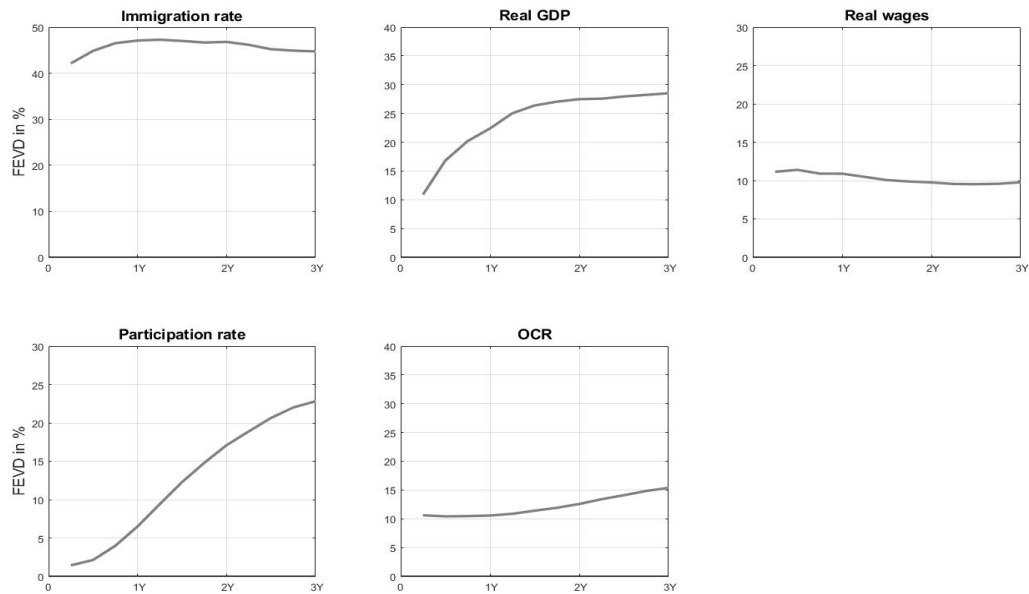


Figure 25: FEVD of the immigration shock using OCR

C.12 FEVD of the immigration shock using 3-Month Interbank Rate

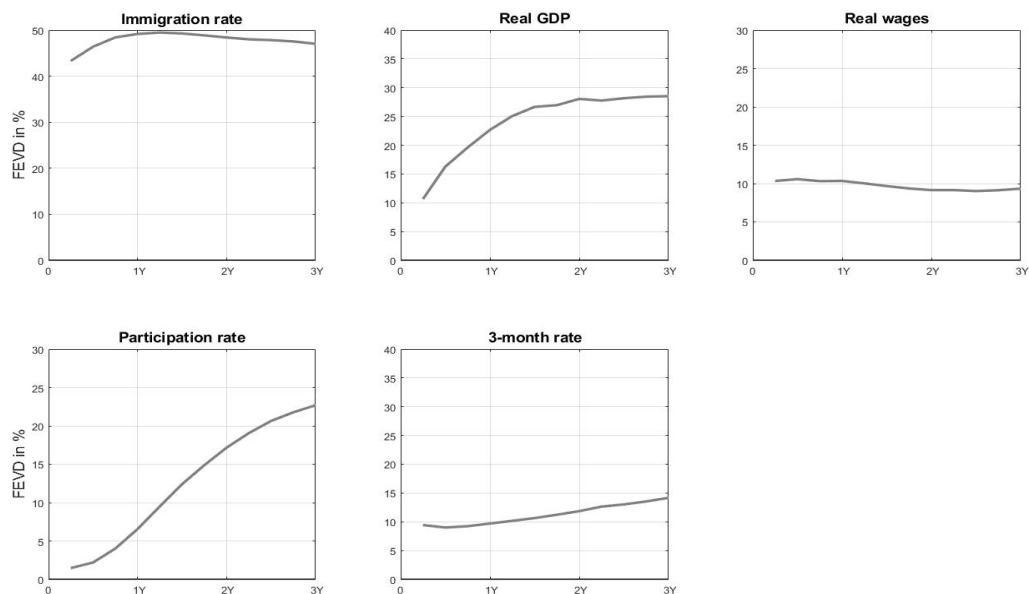


Figure 26: FEVD of the immigration shock using 3-Month Interbank Rate

D FEVD for robustness check - young and old migrants

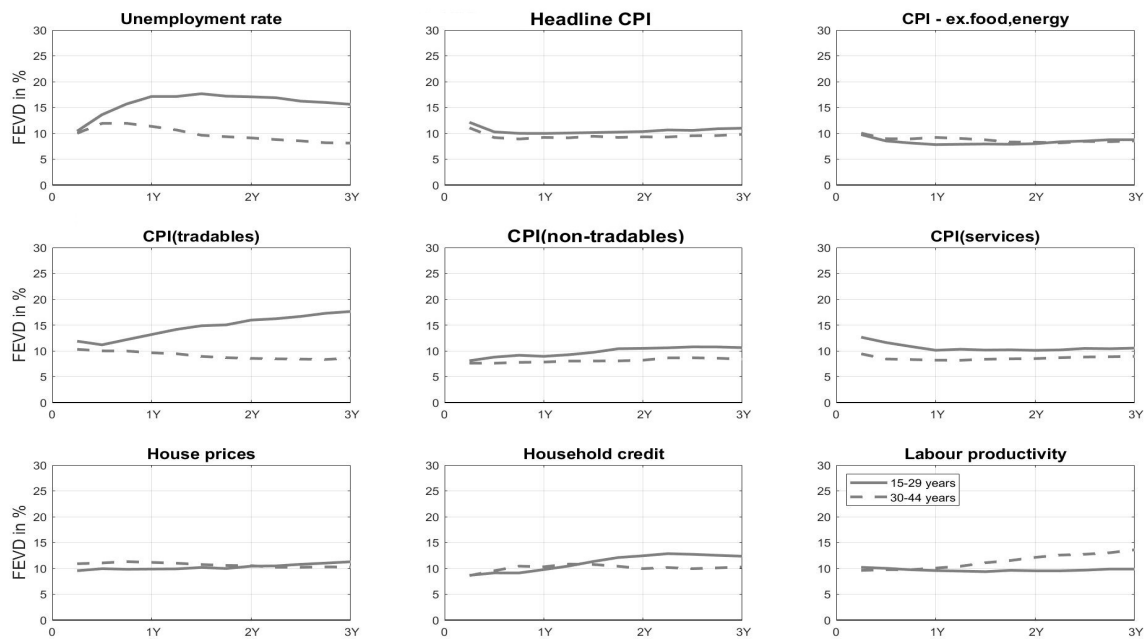


Figure 27: Robustness check: FEVD to the immigration shock. *Note:* The solid (dashed) lines are the point-wise posterior median responses for age 15-29 (30-44).